BOUNDARY REGULARITY FOR THE INHOMOGENEOUS INFINITY LAPLACE EQUATION

Abstract. We extend a boundary regularity argument to the unnormalized equation $-\Delta_\infty u = f$. The starting point is the cited uniform homogeneous interior gradient estimate of the October 4, 2026 OpenAI preprint. Writing $a = a_{n+1}$ for its exponent in one additional dimension, we prove local boundary C^1 regularity for arbitrary $C^{1,\gamma}$ Dirichlet data on $C^{1,\gamma}$ boundary patches, for every $0 < \gamma \le 1$ and every bounded continuous source. The conclusion has a uniform qualitative gradient modulus on bounded classes of charts, traces, solutions, and sources. For a flat patch with constant trace we obtain exponent $a/(2+a)$. For a curved constant-trace patch we obtain $a\lambda/(1+a+\lambda)$, where $\lambda = \min\{\gamma, 1/5\}$. If $f \in C^{0,\sigma}$ vanishes on the patch, the corresponding exponents improve. The proofs use strict anisotropic barriers, a viscosity lift to one additional dimension, and a same-dimensional homogeneous replacement estimate with error controlled by $\|f\|_\infty^{1/5}$. The bounded-source statements hold more generally for two-sided viscosity inequalities and impose no sign condition on the source. For general nonconstant traces, a uniform boundary affine-decay criterion is proved; a spatial Hölder rate without that additional criterion is not claimed.

Draft status. This candidate proof was obtained by our team on October 08, 2026, through extensive dialogue-based training with AI. The team is currently working hard to verify all the proofs.

Contents

1. Introduction 2
1.1. Research background and scope 2
1.2. Main results 3
1.3. Outline of the proof 4
2. Comparison, local geometry, and affine approximation 4
2.1. Viscosity convention and strict comparison 4
2.2. Uniform local graph geometry 5
2.3. Forced radial barriers and ambient Lipschitz continuity 7
2.4. Affine approximation on clipped balls 8
3. Interior approximation for the forced equation 9
4. General boundary data and bounded viscosity inequalities 11
4.1. Asymptotic cone comparison 11
4.2. Boundary differentiability: the two-radius argument 12
4.3. Strict anisotropic barriers 13
4.4. Recentered geometry and the boundary flatness estimate 15
4.5. The general-data boundary theorem 16
5. Flat constant boundary traces 18
6. Homogeneous replacement and constant boundary traces 19
6.1. The gradient-patching input and its error 19
6.2. A quantitative replacement on an arbitrary bounded set 20
6.3. The interior affine estimate needed below 21
6.4. Odd full-ball replacements near a curved zero trace 21
6.5. The curved-boundary Hölder estimate 22
7. A direct transfer from boundary affine decay 24

Date: October 10, 2026.
2020 *Mathematics Subject Classification.* 35J70, 35B65, 35D40.
Key words and phrases. Infinity Laplacian; viscosity solutions; Dirichlet boundary regularity; homogeneous replacement; gradient patching.

8. Calibrated slopes and the range of possible exponents 25
8.1. Forcing changes a calibrated boundary slope 25
8.2. The universal upper obstruction 26
8.3. The additional estimate for general nonconstant data 27
References 27

1. Introduction

Let $\Omega \subset \mathbb{R}^n$, $n \geq 2$, be a domain and consider

$$-\Delta_\infty u = -\langle D^2 u\, Du, Du\rangle = f \quad \text{in } \Omega, \qquad u = g \quad \text{on a boundary patch } \Gamma. \tag{1.1}$$

The operator is the unnormalized infinity Laplacian. In particular, under $v(y) = (u(x_0 + Ry) - c)/A$, its right-hand side becomes $R^4 f(x_0 + Ry)/A^3$. This scaling is essential for every estimate below. We study a prescribed continuous viscosity solution; the argument does not require a uniqueness theorem for a sign-changing inhomogeneous Dirichlet problem.

1.1. Research background and scope.

Jensen's uniqueness theorem and the comparison-with-cones theory of Crandall–Evans–Gariepy provide the homogeneous comparison tools [1, 2]. At the boundary, Wang–Yu proved differentiability in general dimensions under C^1 hypotheses and boundary C^1 regularity in the plane under C^2 hypotheses [4]. Hong proved boundary differentiability for bounded continuous inhomogeneities when the graph and the trace are differentiable at the point in question [5, Theorem 1.4]. That theorem concerns the first-order expansion at one point; a uniform decay rate and continuity of the full gradient are additional questions. Feng–Hong subsequently studied slope estimates and differentiability at corner points of convex domains [7]. These results explain why boundary differentiability and boundary gradient Hölder regularity must be stated separately.

The literature also contains a counterexample to boundary gradient continuity for a merely C^1 planar boundary [6]. The positive result for general data in this paper uses a fixed positive Hölder exponent for both the boundary normal and the tangential derivative of the trace. Smoothness of the domain alone does not restore the exact Lipschitz calibration available for the homogeneous equation; Section 8 gives a smooth forced example showing the change in the normal derivative explicitly.

The homogeneous input used here is the following cited theorem. We use its uniform estimate as an external result; its interior proof is given in the cited preprint. The boundary and forcing estimates proved below are derived from it. The research context is recorded as of October 17, 2026; no priority claim for these deductions is made.

Theorem 1.1 (OpenAI, uniform homogeneous interior estimate)**.** *For every integer $d \geq 3$ there exist $a_d \in (0, 1/3]$ and $C_d < \infty$, depending only on d, such that every bounded infinity-harmonic function $h \in C(B_1^d)$ satisfies*

$$\|Dh\|_{L^\infty(B^d_{1/2})} + [Dh]_{C^{0,a_d}(B^d_{1/2})} \leq C_d \operatorname{osc}_{B_1^d} h. \tag{1.2}$$

This is [9, Theorem 1.1]. Fix henceforth

$$a = a_{n+1}, \qquad C_H = C_{n+1}, \qquad b_0 = \frac{a}{2+a}, \qquad b_\sigma = \frac{a(1+\sigma)}{2+\sigma+a} \quad (0 < \sigma \leq 1). \tag{1.3}$$

An infinity-harmonic function of n variables can be regarded as a function independent of a further variable. Thus the same a and dimension-dependent constants can also be used for n-dimensional homogeneous replacements. The extra dimension is needed by the inhomogeneous lifting argument. For completeness, if $H(y,t) = h(y)$ and a smooth function touches H from above or below, its t-derivative is zero at contact. Its horizontal restriction is a test for h, and the infinity Laplacian of the full test is precisely that of its horizontal restriction. Thus this dimension extension is valid in the viscosity sense.

1.2. **Main results.** We write $\operatorname{osc}_E v = \sup_E v - \inf_E v$ and

$$[v]_{\theta;E} = [v]_{C^{0,\theta}(E)} = \sup_{\substack{x,y\in E\\ x\neq y}} \frac{|v(x)-v(y)|}{|x-y|^\theta}.$$

All boundary statements are local. A $C^{1,\gamma}$ trace means that $G(\xi) = g(\xi, \psi(\xi))$ is $C^{1,\gamma}$ on the full parameter disk of a graph chart $x_n = \psi(x')$. Estimates on a smaller patch depend on its separation from the artificial edges of the larger chart. The notation $C^{1,\beta}$ on a closed patch means that the interior gradient extends to the patch, gives the relative first-order expansion there, and satisfies the full Euclidean pairwise Hölder estimate.

Theorem 1.2 (General Dirichlet traces)**.** *Suppose Γ and $g|_\Gamma$ are $C^{1,\gamma}$, $0 < \gamma \le 1$. If u is continuous up to Γ, with $u = g$ on Γ, and satisfies*

$$-F \le -\Delta_\infty u \le F \tag{1.4}$$

in the viscosity sense in a neighborhood of the patch, for a finite constant $F \ge 0$, then u is C^1 up to every smaller closed patch. On classes with fixed chart buffers and bounded chart and trace $C^{1,\gamma}$ norms, solution oscillation, and F, the extended gradient has a common modulus of continuity. In particular the theorem applies to (1.1) with bounded continuous f.

The modulus in this theorem is qualitative; no power of the distance is asserted for unrestricted nonconstant data.

Theorem 1.3 (Constant traces)**.** *Let u be continuous up to a $C^{1,\gamma}$ patch, with $u = c$ there, and suppose (1.4) holds nearby.*

(i) *On a flat patch, u is C^{1,b_0} up to smaller closed patches.*

(ii) *On a curved patch, u is $C^{1,\beta}$ up to smaller closed patches, where*

$$\lambda = \min\{\gamma, 1/5\}, \qquad \beta = \frac{a\lambda}{1+a+\lambda}. \tag{1.5}$$

(iii) *If the actual source belongs to $C^{0,\sigma}$ up to the patch and vanishes there, the flat exponent improves to b_σ and the curved exponent is given by the same formula with*

$$\lambda = \min\{\gamma, (1+\sigma)/5\}. \tag{1.6}$$

For a patch of radius R, the bounded-source estimates have the form

$$R\|Du\|_{L^\infty(E')} + R^{1+\beta}[Du]_{C^{0,\beta}(E')} \le C(\|u-c\|_{L^\infty(E)} + (R^4F)^{1/3}), \tag{1.7}$$

where E' is a fixed smaller closed patch inside the larger patch E, and β denotes the exponent for the case under consideration. For the source-vanishing improvements one may add $(R^{4+\sigma}[f]_{C^{0,\sigma}(E)})^{1/3}$ to the right-hand side. The constant depends on n, a, C_H, the stated exponents, the rescaled graph norm, and the relative chart buffers.

Theorem 1.4 (Uniform boundary affine decay)**.** *Let Γ be a C^1 patch and let u satisfy (1.4) and a local ambient Lipschitz bound. Suppose there exist $M, r_0 > 0$, $0 < \lambda \le 1$, and vectors p_z, for all boundary centers in a larger patch, such that*

$$\sup_{\overline{\Omega}\cap B_r(z)} |u(x) - g(z) - p_z\cdot(x-z)| \le Mr^{1+\lambda} \qquad (0<r<r_0). \tag{1.8}$$

Then on every smaller closed patch

$$u \in C^{1,\beta}, \qquad \beta = \frac{a\lambda}{1+a+\lambda}. \tag{1.9}$$

The estimate is uniform in the displayed bounds, the local Lipschitz bound, and the chart buffers and geometry.

The theorem isolates the additional estimate required for a general boundary Hölder result. Boundary differentiability by itself gives an $o(r)$ error, not (1.8) with a positive uniform λ. Even when Γ and f are smooth and the trace is constant, exponent $1/3$ is an upper obstruction; the final section gives the explicit one-dimensional profile. The exponents proved here are not claimed to be optimal.

1.3. **Outline of the proof.** Section 2 establishes strict comparison, uniform tangent-coordinate geometry, clipped-ball affine approximation, and forced local Lipschitz bounds. The next section proves the interior estimates by adding an auxiliary variable with a prescribed cubic derivative; its two-scale form is retained for the boundary argument. For general traces, explicit anisotropic barriers convert small boundary affine error and small scaled forcing into gradient closeness. Boundary differentiability and compactness then yield the first main theorem. Odd reflection treats a flat constant trace. For curved constant traces, Jensen's gradient patching gives a homogeneous replacement error of order $F^{1/5}$; an odd full-ball replacement then contributes the geometric error of order $r^{1+\gamma}$. Balancing these errors against the homogeneous interior Taylor estimate proves the stated Hölder exponents. The final argument gives the uniform-affine-decay transfer and explains the limitation of homogeneous chord calibration in the presence of forcing.

2. Comparison, local geometry, and affine approximation

2.1. **Viscosity convention and strict comparison.** For $\mathcal{F}(p, X) = -p^T X p$, the inequality $-\Delta_\infty u \le F$ means $\mathcal{F}(D\phi, D^2\phi) \le F$ at every upper C^2 test for u; the inequality $-\Delta_\infty u \ge -F$ is tested from below. Here F is a pointwise bound, not an almost-everywhere specification of a possibly discontinuous source. We use viscosity stability [3, Section 6] in the following form: if every compact subset of an open set D lies in D_j for all sufficiently large j, $u_j \to u$ locally uniformly there, and $-F_j \le -\Delta_\infty u_j \le F_j$ with $F_j \to F$, then $-F \le -\Delta_\infty u \le F$ in D. The same statement applies to equations whose continuous right-hand sides converge locally uniformly.

Lemma 2.1 (Strict comparison)**.** *Let D be bounded and open, $v, w \in C(\overline{D})$, and $\kappa > 0$. If $-\Delta_\infty v \le F$, $-\Delta_\infty w \ge F + \kappa$ in D and $v \le w$ on ∂D, then $v \le w$ in D.*

Proof. Suppose $m = \max_{\overline{D}}(v - w) > 0$, and let $(x_\varepsilon, y_\varepsilon)$ maximize $v(x) - w(y) - |x - y|^2/(2\varepsilon)$ on $\overline{D} \times \overline{D}$. The maximum is at least m. Boundedness first gives $|x_\varepsilon - y_\varepsilon| \to 0$. Every convergent subsequence of these pairs therefore has a diagonal limit (z, z), and continuity gives $v(z) - w(z) \ge m$. Since $v \le w$ on ∂D, such a z is interior. Compactness thus places both maximizing points in D for all sufficiently small ε.

The theorem of sums [3, Theorem 3.2] supplies closed jets with common gradient $p = (x_\varepsilon - y_\varepsilon)/\varepsilon$ and matrices X, Y satisfying

$$\begin{pmatrix} X & 0 \\ 0 & -Y \end{pmatrix} \le A + \tau A^2, \qquad A = \frac{1}{\varepsilon}\begin{pmatrix} I & -I \\ -I & I \end{pmatrix}, \qquad \tau > 0.$$

Since $A(\xi, \xi) = 0$, evaluation on (ξ, ξ) gives $X \le Y$. The viscosity inequalities extend to these jets because the operator and the constant bounds are continuous. Consequently

$$F \ge -p^T X p \ge -p^T Y p \ge F + \kappa,$$

which is impossible. No bound on the possibly large common gradient is required. □

We also use the classical homogeneous comparison and Dirichlet existence theorems: ordered continuous boundary values order a homogeneous subsolution and supersolution, and every continuous boundary value on a bounded open set has a unique continuous infinity-harmonic extension [1, 2, 5]. We use these results for continuous boundary values, applying them componentwise when the open set is disconnected. Only this homogeneous comparison and the strict comparison just proved will be used. In particular, we do not compare two arbitrary solutions with the same sign-changing inhomogeneity.

2.2. **Uniform local graph geometry.** We make the local neighborhoods precise. In each local argument, work in a master cylinder $V = D \times I$, where $D \subset \mathbb{R}^{n-1}$ is a disk, the physical boundary is the graph $X(\xi) = (\xi, \psi(\xi))$, $X(D) \subset V$, and $\Omega \cap V = \{x \in V : x_n > \psi(x')\}$. The graph and its coordinate trace are defined on D. Choose open neighborhoods $U_0 \Subset U_1 \Subset V$ whose projected closures lie in D. Final conclusions are on $\overline{\Omega} \cap U_0$. Set

$$\begin{aligned} \sigma &= \operatorname{dist}(\overline{U}_0, \mathbb{R}^n \setminus U_1) > 0, \\ b &= \min\{\operatorname{dist}(\overline{U}_1, \mathbb{R}^n \setminus V), \operatorname{dist}(\pi' \overline{U}_1, \mathbb{R}^{n-1} \setminus D)\} > 0. \end{aligned} \tag{2.1}$$

Every subsequent radius is restricted by these separations as well as by the stated chart radius. One can always insert additional intermediate neighborhoods before applying a local estimate. On a compact boundary patch, finitely many such configurations suffice. For a bounded family of charts the separations are chosen uniformly. A Lebesgue number for a finite cover places every sufficiently close pair in one configuration. Pairs with larger separation are controlled by a uniform gradient bound when combining the local conclusions. Throughout, $\partial\Omega$ and $d(x) = \operatorname{dist}(x, \partial\Omega)$ refer to the physical domain, not to the artificial cylinder boundary.

Lemma 2.2 (Geometry of graph patches)**.** *Suppose $\psi \in C^{1,\gamma}(D)$, and put $K = \|D\psi\|_\infty$ and $M = [D\psi]_{C^{0,\gamma}}$. With the neighborhoods above, the following properties hold.*

(i) *The inward normal is $C^{0,\gamma}$ on the graph. Choose $\rho > 0$ so that*

$$\rho \le \frac{b}{16(1+K)}, \qquad M(4\rho)^\gamma \le \frac{1}{16}. \tag{2.2}$$

The second restriction is omitted when $M = 0$. At every $z \in \partial\Omega \cap U_1$ there are orthonormal tangent coordinates $y \in \mathbb{R}^{n-1}$ and an inward normal coordinate t in which the graph is $t = \psi_z(y)$ on the full disk $B'_{2\rho}$. Moreover $\psi_z(0) = 0$, $D\psi_z(0) = 0$, and

$$\|D\psi_z\|_\infty \le \frac{1}{15}, \qquad [D\psi_z]_{C^{0,\gamma}} \le C_K M. \tag{2.3}$$

In the entire rotated cylinder $|y| < \rho$, $|t| < \rho$, the domain is exactly $\{t > \psi_z(y)\}$. If $G \in C^{1,\gamma}(D)$ is the original coordinate trace and G_z the trace in these tangent coordinates, then

$$[DG_z]_{C^{0,\gamma}} \le C_K\big([DG]_{C^{0,\gamma}} + \|DG\|_\infty M\big). \tag{2.4}$$

In addition, $\|DG_z\|_\infty \le (16/15)\|DG\|_\infty$. Writing $N_G = [DG]_{C^{0,\gamma}} + \|DG\|_\infty M$, the following uniform Taylor estimates hold on the full disk:

$$\begin{aligned} |\psi_z(y)| &\le C_K M |y|^{1+\gamma}, \\ |G_z(y) - G_z(0) - DG_z(0)\cdot y| &\le C_K N_G |y|^{1+\gamma}. \end{aligned} \tag{2.5}$$

(ii) *For $x \in \overline{\Omega} \cap U_0$ and every sufficiently small r, depending only on the neighborhood separations, one has*

$$B_{\kappa r}\big(x + \tfrac{r}{2} e_n\big) \subset \Omega \cap B_r(x), \qquad \kappa = \frac{1}{4(1+K)}. \tag{2.6}$$

(iii) *If $x \in \Omega \cap U_0$ and $d(x) < \sigma/4$, every nearest boundary point z belongs to U_1 and satisfies $x - z = d(x)\nu(z)$. For $y \in B_{d(x)/4}(x)$ every nearest boundary point w also belongs to U_1, and*

$$\tfrac{3}{4} d(x) \le d(y) \le \tfrac{5}{4} d(x), \qquad |w - z| \le \tfrac{5}{2} d(x). \tag{2.7}$$

Assertion (ii) *holds already for a Lipschitz graph with $K = \operatorname{Lip}(\psi, D)$. Assertion* (iii) *needs only C^1 regularity. Tangent charts also exist uniformly on smaller C^1 patches, with the second condition in* (2.2) *replaced by the corresponding modulus-of-continuity condition.*

Proof. We use column vectors for gradients. Write $J(\xi) = DX(\xi)$, an $n \times (n-1)$ matrix. The formula $\nu(X(\xi)) = (-D\psi(\xi), 1)/(1 + |D\psi(\xi)|^2)^{1/2}$ and $|\xi - \zeta| \le |X(\xi) - X(\zeta)|$ give the normal estimate.

A common tangent-coordinate disk. Fix $z = X(\xi_0) \in U_1$, let $J_0 = J(\xi_0)$, and set

$$A_0 = (J_0^T J_0)^{1/2}, \qquad P_z = J_0 A_0^{-1}.$$

The columns of P_z are an orthonormal tangent frame, $P_z^T J_0 = A_0$, and $\left\|A_0^{-1}\right\| \le 1$, since $J_0^T J_0 = I + D\psi(\xi_0) \otimes D\psi(\xi_0) \ge I$. The matrix $R_z = (P_z, \nu(z))$ is orthogonal. Define the tangential projection

$$H_z(\xi) = P_z^T(X(\xi) - z).$$

By (2.2), the disk $B'_{4\rho}(\xi_0)$ lies in D. Its graph lies in V, since

$$|X(\xi) - z| \le (1 + K)|\xi - \xi_0| \le 4(1 + K)\rho \le b/4.$$

Moreover,

$$\|DH_z(\xi) - A_0\| \le M(4\rho)^\gamma \le 1/16.$$

Write $H_z(\xi_0 + h) = A_0 h + \mathcal{R}(h)$. Integration along segments in the convex parameter disk shows that $\mathcal{R}(0) = 0$ and $\mathrm{Lip}(\mathcal{R}) \le 1/16$. Consequently

$$|H_z(\xi) - H_z(\zeta)| \ge \tfrac{15}{16}|\xi - \zeta|. \tag{2.8}$$

This rules out multiple graph points with the same projection in the chosen disk.

For $|y| \le 2\rho$, the map $h \mapsto A_0^{-1}(y - \mathcal{R}(h))$ is a contraction of $\overline{B}'_{4\rho}$ into itself: its norm is at most $2\rho + 4\rho/16 < 4\rho$ and its Lipschitz constant is at most $1/16$. Its unique fixed point defines the inverse $\xi = \Xi_z(y)$ on $B'_{2\rho}$. The inverse-function theorem now applies on this already specified disk. It gives

$$D\Xi_z(y) = DH_z(\Xi_z(y))^{-1}, \qquad \|D\Xi_z\| \le 16/15.$$

For invertible matrices B, C, $B^{-1} - C^{-1} = B^{-1}(C - B)C^{-1}$. Together with (2.8), this gives, for $y, v \in B'_{2\rho}$,

$$|D\Xi_z(y) - D\Xi_z(v)| \le (16/15)^2 M|\Xi_z(y) - \Xi_z(v)|^\gamma \le (16/15)^{2+\gamma} M|y - v|^\gamma.$$

Set

$$\psi_z(y) = \nu(z)^T(X(\Xi_z(y)) - z), \qquad G_z(y) = G(\Xi_z(y)).$$

With $\xi = \Xi_z(y)$, the exact derivative formulas are

$$\begin{aligned} D\psi_z(y) &= DH_z(\xi)^{-T} J(\xi)^T \nu(z), \\ DG_z(y) &= DH_z(\xi)^{-T} DG(\xi). \end{aligned} \tag{2.9}$$

Since $J_0^T \nu(z) = 0$, the first gradient is zero at the origin and has norm at most $(16/15)(1/16) = 1/15$. The same inverse-matrix identity and the product rule for Hölder seminorms prove (2.3) and (2.4). All constants are uniform in z. The bound for $\|DG_z\|_\infty$ follows directly from (2.9). Integrating these derivative estimates along $\theta \mapsto \theta y$, $0 \le \theta \le 1$, which remains inside the full convex tangent disk, proves (2.5); the integral of θ^γ is $1/(1 + \gamma)$. This gives the common Taylor constants used in the subsequent boundary constructions.

The side occupied by the domain. For $Y = z + P_z y + t\nu(z)$ with $|y|, |t| < \rho$, set $F_z(y, t) = Y_n - \psi(Y')$. These points stay in V by (2.2), and

$$|Y' - \xi_0| \le |Y - z| = (|y|^2 + t^2)^{1/2} < \sqrt{2}\rho < 4\rho.$$

If $a_0 = D\psi(\xi_0)$, then

$$\partial_t F_z(y, t) = \frac{1 + a_0 \cdot D\psi(Y')}{\sqrt{1 + |a_0|^2}} \ge 1 - M(4\rho)^\gamma \ge 15/16.$$

The point $t = \psi_z(y)$ is a zero, and $|\psi_z(y)| \le |y|/15 < \rho$. Thus the sign of F_z is exactly the sign of $t - \psi_z(y)$. Since $\Omega \cap V$ is the positive set of the original defining function, this proves the asserted representation of the entire local domain.

Thickness. Use the original graph coordinates and take κ as in (2.6). For $Y \in B_{\kappa r}(x + (r/2)e_n)$,

$$Y_n - \psi(Y') \ge x_n - \psi(x') + \tfrac{r}{2} - (1 + K)\kappa r \ge r/4 > 0.$$

Also $|Y-x| \le r/2 + \kappa r \le 3r/4$. Choosing r below the buffer radius keeps this ball in V and proves (2.6). Only the Lipschitz inequality for ψ was used.

Nearest boundary points. The distance to the nonempty closed set $\partial\Omega$ is attained. If $d(x) < \sigma/4$, every minimizer z has $|z-x| = d(x)$, hence belongs to U_1. Differentiating $|x - X(\xi)|^2$ at such a minimizer gives $J(\xi_z)^T(x-z) = 0$. Therefore $x - z = \pm d(x)\nu(z)$. The open segment $(z, x]$ lies in Ω, because all its points other than z lie in $B_{d(x)}(x)$. An initial outward normal segment lies outside Ω, as follows from the positive normal derivative of $Y_n - \psi(Y')$ at z. The minus sign is therefore impossible. This sign argument uses neither a uniform tubular neighborhood nor uniqueness of the minimizer.

Finally, the distance function is 1-Lipschitz. Hence $3d(x)/4 \le d(y) \le 5d(x)/4$ and $|w-x| \le d(y) + |y-x| \le 3d(x)/2 < \sigma$, so every such w also belongs to U_1. The triangle inequality gives $|w-z| \le d(y) + |y-x| + d(x) \le 5d(x)/2$. For a merely C^1 graph, take the modulus of continuity $\omega_{D\psi}$ on a compact enlarged parameter disk and require $\omega_{D\psi}(4\rho) \le 1/16$. The same tangent construction then gives a common radius on every smaller patch. □

2.3. **Forced radial barriers and ambient Lipschitz continuity.**

Lemma 2.3 (Forced local Lipschitz bounds)**.** *Suppose u satisfies (1.4). If $\overline{B}_d(x_0) \subset \Omega$ and $M = \operatorname{osc}_{B_d(x_0)} u$, then*

$$|u(x) - u(x_0)| \le \left((M/d)^3 + 3Fd\right)^{1/3} |x - x_0| \qquad (x \in B_d(x_0)). \tag{2.10}$$

In particular, $\operatorname{osc}_{B_1} u \le 1$, $F \le 1$ imply $\operatorname{Lip}(u; B_{3/4}) \le 9$. If u is continuous up to a Lipschitz graph patch and the coordinate trace is Lipschitz, then u is locally Lipschitz on the closed patch for the ambient Euclidean distance. The bound depends only on the graph and trace Lipschitz bounds, the solution oscillation, F, and the fixed neighborhood buffers.

Proof. For $L_0 \ge 0$, $R > 0$, and $\varepsilon > 0$, put

$$q(s) = \int_0^s \left(L_0^3 + 3(F+\varepsilon)(R-t)\right)^{1/3} dt, \qquad 0 \le s \le R. \tag{2.11}$$

Then $L_0 s \le q(s) \le q'(0)s$, and $-(q')^2 q'' = F + \varepsilon$ for $0 < s < R$. For a radial function, $\Delta_\infty(q(|x-z|)) = (q')^2 q''$ away from the center: the tangential Hessian has no contribution because the gradient is radial.

Take $R = d$ and $L_0 = M/d$. The function $u(x_0) + q(|x - x_0|)$ bounds u on the sphere and agrees with it at the center; continuity extends the oscillation bound to the sphere. Apply Lemma 2.1 on $B_d(x_0) \setminus \{x_0\}$, where the radial function is smooth and its right-hand side exceeds F by ε. The same comparison for $-u$, followed by $\varepsilon \downarrow 0$, proves (2.10). The radicand is positive at every possible interior contact, including when $L_0 = 0$. In the normalized ball, use $d = 1/8$ at every center of $B_{3/4}$. The coefficient is less than 9; subdivision of each segment within this convex ball gives the asserted pairwise bound.

For the boundary assertion choose a common radius $R \le \min\{b, \sigma\}/8$, with the buffers of (2.1). Let T bound the coordinate trace Lipschitz constant, and let M_0 bound the solution oscillation in that chart. The trace Lipschitz constant for ambient distance is at most T, since horizontal projection decreases distance. For each $z \in \partial\Omega \cap U_1$, take $L_0 = T + M_0/R$ in (2.11). On the physical graph the two radial barriers $g(z) \pm q(|x-z|)$ order the trace; on the spherical part of $\partial(\Omega \cap B_R(z))$ they order u by $q(R) \ge M_0$. Lemma 2.1 on $\Omega \cap B_R(z)$, followed by $\varepsilon \downarrow 0$, therefore gives

$$|u(x) - g(z)| \le L_b |x - z| \quad (x \in \overline{\Omega} \cap B_R(z)), \qquad L_b = (L_0^3 + 3FR)^{1/3}. \tag{2.12}$$

We explain why this boundary-centered bound gives an ambient pairwise Lipschitz estimate. If $x \in \Omega \cap U_0$ and $0 < d = d(x) < R/8$, take any nearest boundary point z. For $y \in B_{d/2}(x)$, (2.12) gives $|u(y) - g(z)| \le (3/2)L_b d$; hence the oscillation on that ball is at most $3L_b d$. Apply (2.10) directly at x with radius $d/2$. Since $(Fd)^{1/3} \le (FR)^{1/3} \le L_b$, it gives

$$|u(y) - u(x)| \le C L_b |y - x| \qquad (y \in B_{d/2}(x)).$$

Because $R \le \sigma/8$, set $h_0 = R/40$. For a pair x, y in the final closed patch with $h = |x-y| < h_0$ and $h \ge d(x)/4$, choose nearest points z, w for x, y. The distance function is 1-Lipschitz, so $d(x) \le 4h$, $d(y) \le 5h < R/8$, and $|z-w| \le 10h$. Both feet belong to U_1, and both radius-R cone estimates apply. Using (2.12) twice and the trace Lipschitz bound gives $|u(x)-u(y)| \le CL_b h$. The same argument includes boundary centers by taking their distance to be zero. If $h < d(x)/4$ and $d(x) < R/8$, the preceding interior-ball argument applies. If $h < d(x)/4$ and $d(x) \ge R/8$, the segment from x to y is interior, and every point ζ of it satisfies

$$d(\zeta) \ge d(x) - h > R/8 - R/40 = R/10.$$

Radius-$R/32$ interior radial estimates along the segment, and subdivision if needed, give the bound $C(M_0/R + (FR)^{1/3}) \le CL_b$. These balls remain in U_1, since their points lie within $h + R/32 < \sigma$ of $x \in U_0$. For $h \ge h_0$, use $|u(x) - u(y)| \le M_0 \le 40 L_b h$. These cases prove the assertion, without requiring a unique nearest point or a segment within the domain for a distant pair. □

2.4. **Affine approximation on clipped balls.**

Lemma 2.4 (Clipped-ball affine criterion)**.** *Let E be the part of $\overline{\Omega}$ in a larger graph neighborhood, and let E' be a smaller closed patch. Suppose, uniformly for $x \in E'$ and $0 < r \le r_*$, that $E \cap B_r(x)$ contains an ordinary ball of radius κr, and that an affine function $P_{x,r}$ satisfies*

$$P_{x,r}(x) = u(x), \qquad |DP_{x,r}| \le K, \qquad \sup_{E \cap B_r(x)} |u - P_{x,r}| \le M r^{1+b}, \quad 0 < b < 1. \tag{2.13}$$

Then the slopes converge to a field $p(x)$ with $|p| \le K$,

$$|u(y) - u(x) - p(x) \cdot (y-x)| \le C_{b,\kappa} M |y-x|^{1+b} \tag{2.14}$$

for $y \in E$ sufficiently close to x, and

$$[p]_{C^{0,b}(E')} \le C_{b,\kappa}(M + K r_*^{-b}).$$

In the interior $p = Du$, and this is the relative gradient at a boundary point. The same statement holds for full interior balls.

Proof. An affine function with absolute value at most N on a ball of radius t has gradient norm at most N/t: evaluate it at opposite ends of each diameter and take the supremum over directions. Thus, whenever $r/2 \le s \le r \le r_*$, subtracting the two approximations on $E \cap B_s(x)$ and using its thickness ball gives

$$|DP_{x,r} - DP_{x,s}| \le \frac{M(r^{1+b} + s^{1+b})}{\kappa s} \le C_b \kappa^{-1} M r^b.$$

Apply this single comparison first to successive dyadic radii $r_j = 2^{-j} r_*$. Summation of the geometric series gives a limit $p(x)$, with $|p(x)| \le K$, and $|DP_{x,r_j} - p(x)| \le C_{b,\kappa} M r_j^b$. For any $0 < r \le r_*$ choose j with $r_j/2 < r \le r_j$ and apply the same comparison to r_j, r. Together with the dyadic tail, this proves $|DP_{x,r} - p(x)| \le C_{b,\kappa} M r^b$ for every radius. In particular, the full family of slopes has the same limit. Inserting the dyadic tail in (2.13) at a dyadic radius with $|y-x| < r_j \le 2|y-x|$, for $0 < |y-x| < r_*/2$, proves (2.14). The strict inequality keeps the evaluation point in the open comparison ball. The derivative in an interior ball is thereby identified. If $p(x)$ and $q(x)$ give two relative first-order expansions at a boundary point, their difference obeys

$$\sup_{z \in E \cap B_r(x)} |(p(x) - q(x)) \cdot (z - x)| = o(r).$$

The affine gradient bound on the thickness ball then gives $|p(x) - q(x)| \le o(r)/(\kappa r) \to 0$, proving uniqueness.

For $x, y \in E'$ with $0 < h = |x-y| \le r_*/4$, compare their limiting affine expansions on $E \cap B_h(y) \subset B_{2h}(x)$. Their difference is bounded there by $C_{b,\kappa} M h^{1+b}$. The thickness ball of radius κh bounds the difference of the slopes by $C_{b,\kappa} M h^b$. For larger separations use

$|p(x) - p(y)| \le 2K$. This proves the full pairwise estimate. Full interior balls are already the special case $\kappa = 1$. □

3. Interior approximation for the forced equation

Throughout this section, $a = a_{n+1} \in (0, 1/3]$ denotes an exponent in Theorem 1.1, in dimension $n+1$. We use its scaled unit-ball formulation

$$\|Dh\|_{L^\infty(B^{n+1}_{1/2})} + [Dh]_{a;B^{n+1}_{1/2}} \le C_H \operatorname{osc}_{B^{n+1}_1} h \tag{3.1}$$

for infinity-harmonic h. The dimension increase is essential to the construction below. All inequalities use the pointwise viscosity convention of Section 2.

Lemma 3.1 (Smooth lifting and homogeneous comparison)**.** *Suppose $v \in C(G)$ satisfies $q_-(y) \le -\Delta_\infty v \le q_+(y)$ in an open set $G \subset \mathbb{R}^n$. If I is an open interval and $w \in C^2(I)$, then $V(y,t) = v(y) + w(t)$ satisfies*

$$q_-(y) - (w')^2 w'' \le -\Delta_\infty V \le q_+(y) - (w')^2 w'' \qquad \text{in } G \times I. \tag{3.2}$$

For $|c| \le 2$, $|t| \le 1$, put

$$W_c(t) = \int_0^t (8 + 3cs)^{1/3}\, ds. \tag{3.3}$$

If $v \in C(\overline{B^n_1})$ satisfies $c - \delta \le -\Delta_\infty v \le c + \delta$, with $\delta \ge 0$ and $|c| + \delta \le 2$, let h be the homogeneous Dirichlet solution in B^{n+1}_1 with boundary values $V(y,t) = v(y) + W_c(t)$. Then

$$\|h - V\|_{L^\infty(B^{n+1}_1)} \le \delta, \qquad \|h(\cdot, 0) - v\|_{L^\infty(B^n_1)} \le \delta. \tag{3.4}$$

Proof. Let ϕ be an upper test for V at (y_0, t_0), and write $p = D_y\phi(y_0, t_0)$ and $k = w'(t_0)$. Vertical contact gives $\phi_t = k$. If $p \ne 0$, set $b = kp/|p|^2$ and consider the upper test for v

$$\psi(y) = \phi(y, t_0 + b \cdot (y - y_0)) - w(t_0 + b \cdot (y - y_0)).$$

It is defined in a sufficiently small neighborhood of y_0, which is all that a viscosity test requires. At the contact point, $D\psi = p$ and, writing $H = D^2_{yy}\phi$, $q = D_y\phi_t$, and $\eta = \phi_{tt}$,

$$D^2\psi = H + q \otimes b + b \otimes q + (\eta - w'')b \otimes b.$$

Since $p \cdot b = k$, this identity includes the mixed terms and yields

$$p^T D^2\psi p = (p,k)^T D^2\phi (p,k) - k^2 w''.$$

The upper-test inequality for v proves the upper inequality in (3.2). If $p = 0$, the horizontal slice is an upper test for v with zero gradient, so $0 \le q_+(y_0)$; the vertical slice gives $\phi_{tt} \ge w''$. Hence

$$-\Delta_\infty \phi = -k^2 \phi_{tt} \le -k^2 w'' \le q_+(y_0) - k^2 w''.$$

Apply the upper-test argument just proved to $-v$ and $-w$, using the upper bound $-q_-$ for $-\Delta_\infty(-v)$. Since the infinity Laplacian is odd under multiplication by -1, negating the resulting inequality gives the lower bound in (3.2), including the case of zero horizontal gradient.

The radicand in (3.3) lies between 2 and 14; in particular W_c is smooth on a neighborhood of $[-1, 1]$. Direct differentiation gives

$$(W_c')^2 W_c'' = c, \qquad W_c(0) = 0, \qquad \partial_c W_c(t) = t^2 \int_0^1 \tau (8 + 3ct\tau)^{-2/3}\, d\tau \in [0, 1].$$

The expression with the factor t^2 also proves these bounds when $t < 0$. Since $[c - \delta, c + \delta] \subset [-2, 2]$, the mean value theorem in the parameter gives, uniformly for $|t| \le 1$, $0 \le W_{c+\delta} - W_c \le \delta$ and $0 \le W_c - W_{c-\delta} \le \delta$. Define

$$S = v + W_{c+\delta} - \delta, \qquad T = v + W_{c-\delta} + \delta.$$

The lifting just proved shows that $-\Delta_\infty S \le 0$ and $-\Delta_\infty T \ge 0$. On the entire closed ball,

$$V - \delta \le S \le V \le T \le V + \delta.$$

The homogeneous Dirichlet comparison principle consequently gives $S \le h \le T$, proving (3.4). Only homogeneous comparison is used; no comparison principle for sign-changing inhomogeneous sources is required. □

Proposition 3.2 (Two-scale affine approximation)**.** *Suppose $u \in C(B_1)$, $\operatorname{osc}_{B_1} u \le 1$, and $-1 \le -\Delta_\infty u \le 1$. For every $x \in B_{1/2}$ and $0 < r \le 1/8$, there is $p_{x,r} \in \mathbb{R}^n$, with $|p_{x,r}| \le C$, such that, for $0 < \rho \le r/2$,*

$$\sup_{|z|\le\rho} |u(x+z) - u(x) - p_{x,r}\cdot z| \le C(r^2 + \rho^{1+a} r^{-a}). \tag{3.5}$$

If, in addition, $-\Delta_\infty u = f$, $0 < \sigma \le 1$, and $\|f\|_\infty + [f]_{\sigma;B_1} \le 1$, the first error improves to

$$\sup_{|z|\le\rho} |u(x+z) - u(x) - p_{x,r}\cdot z| \le C(r^{2+\sigma} + \rho^{1+a} r^{-a}). \tag{3.6}$$

Here and below constants may depend on n, a, C_H.

Proof. Since $x + r\overline{B}_1 \subset B_{3/4}$, Lemma 2.3 gives $|u(x+ry) - u(x)| \le 9r|y|$. Thus $v(y) = (u(x+ry) - u(x))/r$ is continuous on $\overline{B}_1$, has $v(0) = 0$ and $|v| \le 9$, and has viscosity source bounds $\pm r$. Apply Lemma 3.1 with $c = 0$ and $\delta = r$. In the Hölder case, the rescaled source is $rf(x+ry)$; use $c = rf(x)$ and $\delta = r^{1+\sigma}[f]_{\sigma;B_1}$. In the Hölder case, $|c| + \delta \le r(\|f\|_\infty + [f]_{\sigma;B_1}) \le r$, so the lifting hypothesis $|c| + \delta \le 2$ holds in both cases. By homogeneous comparison with constants, the oscillation of h is at most the oscillation of its boundary values $v + W_c$, hence at most $18 + 2\cdot 14^{1/3}$. Thus (3.1) bounds its gradient and its a-Hölder seminorm on $B^{n+1}_{1/2}$ by a fixed constant. Take $p_{x,r} = D_y h(0,0)$. Integration along the horizontal segment gives

$$|h(y,0) - h(0,0) - p_{x,r}\cdot y| \le C|y|^{1+a} \qquad (|y| \le 1/2).$$

At $|y| = 1/2$ this follows by continuity from the open half-ball. Since $W_c(0) = 0$ and $v(0) = 0$, the comparison error is used at both $(y,0)$ and $(0,0)$:

$$|v(y) - p_{x,r}\cdot y| \le 2\delta + C|y|^{1+a}.$$

Multiplication by r and substitution $y = z/r$ prove both claims. The slope bound follows from (3.1). No equation is asserted for the horizontal slice of h. □

Theorem 3.3 (Interior estimates with forcing)**.** *Use the exponents b_0, b_σ defined in (1.3). Let $u \in C(B_R)$ be bounded and let $F \ge 0$. If u satisfies $-F \le -\Delta_\infty u \le F$ in B_R, then*

$$R\|Du\|_{L^\infty(B_{R/2})} + R^{1+b_0}[Du]_{b_0;B_{R/2}} \le C\left(\operatorname{osc}_{B_R} u + (R^4 F)^{1/3}\right). \tag{3.7}$$

If $-\Delta_\infty u = f \in C^{0,\sigma}(B_R)$ for $0 < \sigma \le 1$, then

$$\begin{aligned} R\|Du\|_{L^\infty(B_{R/2})} + R^{1+b_\sigma}[Du]_{b_\sigma;B_{R/2}} \le C\big(&\operatorname{osc}_{B_R} u \\ &+ (R^4\|f\|_\infty + R^{4+\sigma}[f]_{\sigma;B_R})^{1/3}\big). \end{aligned} \tag{3.8}$$

The constants in the second assertion may also depend on σ.

Proof. First take $R = 1$ and normalize by $A = \operatorname{osc} u + F^{1/3}$ in the first case, or by $A = \operatorname{osc} u + (\|f\|_\infty + [f]_\sigma)^{1/3}$ in the second. If $A = 0$, the conclusion is immediate. Otherwise replace u by $(u - u(0))/A$. Its oscillation is at most one, and its forcing bound, or the sum of its source sup norm and Hölder seminorm, is at most one because these quantities are divided by A^3. We retain the notation u for this normalized function.

The two cases can now be treated together. Put $m = 2$ for bounded inequalities, and $m = 2+\sigma$ for a Hölder source. The preceding proposition gives errors of the form $C(r^m + \rho^{1+a} r^{-a})$. Set

$$\theta = \frac{1+a}{m+a}, \qquad r = \rho^\theta, \qquad b = \frac{a(m-1)}{m+a}.$$

Thus $b = b_0$ or $b = b_\sigma$, respectively, and

$$r^m = \rho^{1+a} r^{-a} = \rho^{1+b}.$$

Since $2 \le m \le 3$ and $0 < a \le 1/3$, we have $1/3 \le \theta \le 2/3$. Thus, for $0 < \rho \le \rho_* := 2^{-9}$, $r \le 1/8$ and $\rho/r = \rho^{1-\theta} \le 1/8$, so every radius required by the proposition is admissible. For each center $x \in B_{1/2}$ and each such ρ, set $P_{x,\rho}(y) = u(x) + p_{x,r} \cdot (y - x)$. Its slope is uniformly bounded and its error on $B_\rho(x)$ is at most $C\rho^{1+b}$. Apply the full-ball case of Lemma 2.4 with $E = B_1$, $E' = \overline{B}_s$ for $0 < s < 1/2$, and $r_* = \rho_*$. Here $B_\rho(x) \subset B_1$, so the thickness constant is one. The lemma gives the gradient bound and pairwise b-Hölder bound with constants independent of s. Every pair in $B_{1/2}$ belongs to some $\overline{B}_s$; letting $s \uparrow 1/2$ proves $u \in C^{1,b}(B_{1/2})$ with the asserted uniform estimate. Restoring the amplitude multiplies both gradient norms by A.

For a general ball $B_R(x_*)$, apply the unit-ball estimates to $U(y) = u(x_* + Ry)$. In the inequality case its bounds are $-R^4 F \le -\Delta_\infty U \le R^4 F$. In the Hölder case its source is $f_R(y) = R^4 f(x_* + Ry)$, with

$$\|f_R\|_\infty = R^4 \|f\|_\infty, \qquad [f_R]_{\sigma;B_1} = R^{4+\sigma}[f]_{\sigma;B_R}.$$

Finally, $DU(y) = RDu(x_* + Ry)$ and $[DU]_{b;B_{1/2}} = R^{1+b}[Du]_{b;B_{R/2}}$, which give both displayed scaled estimates. □

4. General boundary data and bounded viscosity inequalities

Throughout this section put $b = b_0 = a/(2+a)$. Theorem 3.3 supplies the following interior estimate: if v satisfies $-F \le -\Delta_\infty v \le F$ in an interior ball B_r, then

$$\|Dv\|_{L^\infty(B_{r/2})} + r^b [Dv]_{C^{0,b}(B_{r/2})} \le C_I \left(r^{-1} \operatorname{osc}_{B_r} v + (rF)^{1/3} \right). \tag{4.1}$$

We use the pointwise viscosity convention of Section 2; no prescribed continuous right-hand side is needed here. All local neighborhoods have their closures in a fixed larger graph neighborhood; radii below are chosen smaller than its distance to the artificial boundary.

4.1. Asymptotic cone comparison.

For $x \in \overline{\Omega}$ and an admissible radius r, define

$$\Sigma_r(x) = (\partial\Omega \cap \overline{B}_r(x)) \cup (\overline{\Omega} \cap \partial B_r(x)).$$

This set contains the boundary required for comparison on the clipped open ball and includes every spherical contact point. Set

$$S_r^\pm(x) = \max\left\{0, \sup_{y \in \Sigma_r(x)\setminus\{x\}} \frac{\pm(u(y) - u(x))}{|y - x|}\right\}, \qquad S_r(x) = \max\{S_r^+(x), S_r^-(x)\}. \tag{4.2}$$

These numbers are finite at interior centers. They are also finite at boundary centers when the trace is locally Lipschitz.

Lemma 4.1 (Asymptotic cone comparison)**.** *Suppose $u \in C(\overline{\Omega})$ satisfies $-F \le -\Delta_\infty u \le F$ locally. Whenever the quantities in (4.2) are finite,*

$$\pm(u(y) - u(x)) \le (S_r^\pm(x) + (3Fr)^{1/3})|y - x|, \qquad y \in \overline{\Omega} \cap B_r(x), \tag{4.3}$$

$$S_\rho^\pm(x) \le S_r^\pm(x) + (3Fr)^{1/3}, \qquad 0 < \rho < r. \tag{4.4}$$

Consequently the limits $S^\pm(x) = \lim_{r\downarrow 0} S_r^\pm(x)$ and $S(x) = \lim_{r\downarrow 0} S_r(x)$ exist and are finite. At interior points $S^+(x) = S^-(x) = S(x) = |Du(x)|$.

Proof. Set $S = S_r^+(x)$ and use the radial profile q from (2.11), with $L_0 = S$, $R = r$, and $\varepsilon = \epsilon > 0$. The properties proved in Lemma 2.3 give

$$St \le q(t) \le q'(0)t, \qquad q'(0) = \left(S^3 + 3(F + \epsilon)r\right)^{1/3} \le S + (3(F + \epsilon)r)^{1/3}.$$

The boundary ordering for $u(x) + q(|\cdot - x|)$ follows from $q(t) \ge St$; the center itself has equality. Apply Lemma 2.1 on the clipped ball, punctured at x if x is interior, where this profile has right-hand side $F + \epsilon$. This proves the upper inequality. Let $\epsilon \downarrow 0$ and apply the same argument to $-u$. Taking suprema proves (4.4).

For either one-sided slope, let $r_j \downarrow 0$ realize its lower limit. Inequality (4.4), for every $0 < \rho < r_j$, bounds its upper limit by $\lim_j (S^{\pm}_{r_j}(x) + (3Fr_j)^{1/3})$. This proves convergence; an outer radius bounds the limit from above. For interior centers, the differentiability supplied by (4.1) identifies both limits with $|Du(x)|$. □

Lemma 2.3 supplies the uniform ambient Lipschitz bound on smaller graph neighborhoods used below.

4.2. **Boundary differentiability: the two-radius argument.** For a prescribed bounded continuous source, boundary differentiability is the theorem of Hong [5, Theorem 1.4]. We prove the version for bounded inequalities because this class is closed under the compactness argument below.

Proposition 4.2 (Boundary differentiability for bounded inequalities)**.** *Suppose the boundary is a C^1 graph and its trace is C^1 in a neighborhood of z_0. If $u \in C(\overline{\Omega})$ satisfies $-F \le -\Delta_\infty u \le F$ there, then there is a unique p_{z_0} such that*

$$u(x) = u(z_0) + p_{z_0} \cdot (x - z_0) + o(|x - z_0|), \qquad x \in \overline{\Omega}, \quad x \to z_0. \tag{4.5}$$

Proof. Translate and rotate so that $z_0 = 0$, the inward normal is e_n, and the graph is $X(\xi) = (\xi, \psi(\xi))$, with $\psi(0) = D\psi(0) = 0$. Subtract the constant $u(0)$. Write $G(\xi) = u(X(\xi))$, $p = DG(0)$, and $S_0 = S(0)$. Differentiating the trace at the origin gives $|p| \le S_0$.

We first prove the precise estimate

$$\limsup_{\substack{x \to 0 \\ x \in \Omega}} |Du(x)| \le S_0. \tag{4.6}$$

For a fixed admissible $r > 0$, compactness of the spherical portions and continuity of u imply

$$\limsup_{\substack{x \to 0 \\ x \in \overline{\Omega}}} \sup_{y \in \overline{\Omega} \cap \partial B_r(x)} \frac{|u(y) - u(x)|}{r} \le \sup_{y \in \overline{\Omega} \cap \partial B_r(0)} \frac{|u(y) - u(0)|}{r} \le S_r(0). \tag{4.7}$$

Indeed, otherwise choose centers tending to zero and points on their spheres violating the inequality; a convergent subsequence of these points lies on $\overline{\Omega} \cap \partial B_r(0)$ and contradicts continuity. Empty spherical portions cause no difficulty.

Fix $\epsilon > 0$. Choose $R > 0$ so small that the closed ball $\overline{B}_{4R}$ lies in the graph neighborhood, the coordinate disk B'_{4R} is available, and

$$\sup_{B'_{4R}} |DG| \le |p| + \epsilon \le S_0 + \epsilon, \quad S_t(0) \le S_0 + \epsilon \ (0 < t \le R), \quad (3FR)^{1/3} \le \epsilon. \tag{4.8}$$

For graph points $z, w \in B_{4R}$, integration along the segment of their parameter coordinates, inside the convex disk B'_{4R}, gives

$$|g(z) - g(w)| \le (S_0 + \epsilon)|z' - w'| \le (S_0 + \epsilon)|z - w|. \tag{4.9}$$

By (4.7) at radius R, choose $0 < \rho_1 < R/8$ so that, for every $z \in \overline{\Omega} \cap B_{2\rho_1}$, the spherical supremum in (4.7) is at most $S_0 + 2\epsilon$. If z is on the true boundary, its graph portion is controlled by (4.9); all involved graph points lie in B_{2R}. Consequently $S_R(z) \le S_0 + 2\epsilon$ for $z \in \partial\Omega \cap B_{2\rho_1}$. Lemma 4.1 therefore gives

$$|u(x) - g(z)| \le (S_0 + 3\epsilon)|x - z|, \quad z \in \partial\Omega \cap B_{2\rho_1}, \quad x \in \overline{\Omega} \cap B_{\rho_1}. \tag{4.10}$$

The radius condition is legitimate because $|x - z| \le 3\rho_1 < R$.

Now fix the second radius $r = \rho_1/4$. Again by (4.7), choose $0 < \rho_2 < r/4$ so that the spherical supremum centered at any $x \in \overline{\Omega} \cap B_{\rho_2}$, at this fixed radius r, is at most $S_0 + 2\epsilon$. Every point w on the true boundary portion of $B_r(x)$ satisfies $|w| \le r + \rho_2 < 2\rho_1$; also $x \in B_{\rho_1}$. Applying (4.10) with center w bounds its quotient by $S_0 + 3\epsilon$. The graph and spherical portions together give $S_r(x) \le S_0 + 3\epsilon$. Thus

$$S(x) \le S_r(x) + (3Fr)^{1/3} \le S_0 + 4\epsilon, \qquad x \in \overline{\Omega} \cap B_{\rho_2}.$$

At interior points $S(x) = |Du(x)|$, proving (4.6) as $\epsilon \downarrow 0$.

The local ambient Lipschitz bound of Lemma 2.3 makes the first-order rescalings precompact on the varying closed domains; one can apply Arzelà–Ascoli to their Lipschitz extensions to common balls. More explicitly, for any sequence $r_j \downarrow 0$ set $u_j(y) = u(r_j y)/r_j$ and $\Omega_j = r_j^{-1}\Omega$. A diagonal subsequence of the extensions converges on every compact ball. The rescaled graphs $\psi(r_j y')/r_j$ converge in C^1 on compact parameter disks to zero, while $G(r_j y')/r_j \to p\cdot y'$ uniformly on those disks. Every limit U is defined on $\overline{H}$, where $H = \{x_n > 0\}$, has $U(0) = 0$, and has trace $U(x', 0) = p \cdot x'$. It is S_0-Lipschitz on $\overline{H}$: any fixed segment in H lies in the rescaled domains for all sufficiently small scales, and (4.6) bounds the gradients along that segment by $S_0 + o(1)$. Continuity includes boundary endpoints.

If $S_0 = |p| > 0$, compare $U(x)$ with its values at the boundary points $\pm t(p/|p|, 0)$ and let $t \to \infty$. The two Lipschitz inequalities give $U(x) = p \cdot x'$. If $S_0 = 0$, this conclusion is immediate.

Suppose $S_0 > |p|$. First assume $S^+(0) = S_0$. For each fixed $R > 0$, the positive slope of u_j at radius R is exactly $S^+_{Rr_j}(0)$, which tends to S_0. The graph quotients at these scales are at most $|p| + o(1)$: write the trace difference as the integral of DG along the parameter segment and use that horizontal projection decreases distance. Since $S_0 > |p|$, for all sufficiently large j the supremum defining this slope is a maximum on the compact spherical portion. Choose a maximizing point y_j there. Passing to a subsequence for this fixed R, the points y_j converge to Re_R on the closed upper hemisphere. Uniform convergence of the Lipschitz extensions, together with $u_j(y_j) = RS^+_{Rr_j}(0)$, gives

$$|e_R| = 1, \quad (e_R)_n \geq 0, \qquad U(Re_R) = S_0 R.$$

Equality in the Lipschitz bound forces $U(te_R) = S_0 t$ for $0 \leq t \leq R$. Choose $R_j \to \infty$ and a convergent subsequence of e_{R_j}. Its limit e satisfies $U(te) = S_0 t$ for every $t \geq 0$ and $e_n > 0$; otherwise the boundary trace at e would give $S_0 \leq |p|$. For any $x \in H$,

$$U(x) \geq S_0 t - S_0 |te - x| \longrightarrow S_0 e \cdot x.$$

Applying this inequality at the boundary points $(x', 0)$ and $(-x', 0)$ gives $p = S_0 e'$. Let $q = x - (x_n/e_n)e \in \partial H$. The Lipschitz upper bound gives

$$U(x) \leq p \cdot q' + S_0 x_n/e_n = S_0 e \cdot x.$$

Therefore $U(x) = p \cdot x' + \sqrt{S_0^2 - |p|^2}\, x_n$. If $S^-(0) = S_0$, applying the same argument to $-u$ gives the negative square root. In the case $S_0 > |p|$, choose once and for all a sign attaining $S_0 = \max\{S^+(0), S^-(0)\}$. The corresponding classification applies to every blowup and gives the same affine function. If both signs attained $S_0 > |p|$, applying the two classifications to one blowup would give opposite nonzero normal components, which is impossible. Thus in every case there is a fixed vector p_0, with $p_0 = (p, 0)$ when $S_0 = |p|$, such that every blowup is $U(y) = p_0 \cdot y$.

To pass from subsequences to (4.5), suppose that $|u(x_j) - p_0 \cdot x_j|/|x_j|$ stays bounded away from zero for some $x_j \to 0$ in $\overline{\Omega} \setminus \{0\}$. Set $r_j = |x_j|$. After passing to a subsequence, the Lipschitz extensions of $u(r_j \cdot)/r_j$ converge locally uniformly to the affine function just identified, and x_j/r_j converges to a point on the closed unit hemisphere. Evaluation there contradicts the chosen errors. Finally, two proposed boundary slopes would have the same scalar products with every strictly inward direction, so they coincide. □

4.3. **Strict anisotropic barriers.**

Lemma 4.3 (A quantitative strict barrier family)**.** *Fix $1 < q \leq 2$. There are positive constants $\mu, k, \delta_0, c_q, C_q$, depending only on q, with the following property. Let $0 < \delta < \delta_0$, $v = mw$, $|w| = 1$ and $\delta/2 \leq m \leq 3$. Put*

$$\lambda = (\mu\delta/m)^2, \quad K = k\mu\delta = km\sqrt{\lambda}, \quad A = \frac{4}{q-1} K \lambda^{q/2},$$

$$z = w \cdot x, \quad \xi = P_{w^\perp} x, \quad S = (|\xi|^2 + \lambda z^2)^{1/2}.$$

and define $B_v^{\pm} = mz \pm KS^q \mp A|z|^q$. *Then in* B_1, *in the viscosity sense,*

$$-\Delta_\infty B_v^+ \geq c_q\delta^3, \qquad -\Delta_\infty B_v^- \leq -c_q\delta^3, \qquad |DB_v^{\pm} - v| \leq C_q\delta. \tag{4.11}$$

Proof. Choose $0 < \mu \leq 1/2$ so small that $\frac{4}{q-1}(8\mu)^q \leq 1/4$. Then $\lambda \leq 1$. For $Q = \operatorname{diag}(I, \lambda)$,

$$D^2(S^q) = qS^{q-2}Q + q(q-2)S^{q-4}(Qx)\otimes(Qx) \leq qS^{q-2}Q$$

away from the origin, and $|D(S^q)| \leq q$ on B_1. Since $K = O_q(\delta)$ and $A \leq C_q\delta$, the gradient bound follows. For $P = DB_v^+$ and $z \neq 0$,

$$P_\xi = KqS^{q-2}\xi, \qquad P_z = m + Kq\lambda zS^{q-2} - Aq|z|^{q-2}z.$$

The inequality $S \geq \sqrt{\lambda}|z|$ gives $|P_z - m| \leq C_qK\lambda^{q/2}$. Dividing by m bounds this error by $C_qk\mu^{q+1}(\delta/m)^{q+1}$, which is at most $1/2$ after choosing k small, since $\delta/m \leq 2$. Choose k small enough also that

$$q^2k^2 \leq 1/16, \qquad \frac{16}{q-1}k\mu^{q+1}2^{q-1} \leq 1/4,$$

and fix $0 < \delta_0 \leq 1/2$. In particular $P_z \geq m/2$. The preceding Hessian bound gives

$$\Delta_\infty B_v^+ \leq q^3K^3S^{3q-4} - 3qK\lambda^{q/2}|z|^{q-2}P_z^2.$$

For the entire interval $1 < q \leq 2$ one has

$$S^{3q-4}|z|^{2-q}\lambda^{1-q/2} = S^{3q-4}(\sqrt{\lambda}|z|)^{2-q} \leq S^{2q-2} \leq 1.$$

Consequently, using $q^2k^2 \leq 1/16$,

$$\begin{aligned} -\Delta_\infty B_v^+ &\geq qK\lambda^{q/2}|z|^{q-2}(3m^2/4 - q^2K^2/\lambda) \\ &\geq c_q\delta^{q+1}m^{2-q} \geq c_q\delta^3. \end{aligned}$$

Here $|z|^{q-2} \geq 1$, $K^2/\lambda = k^2m^2$, and $m \geq \delta/2$.

If $q < 2$, at a point $(\xi_0, 0)$ with $\xi_0 \neq 0$ the restriction to the w direction is $B_v^+(\xi_0, 0) + mh - A|h|^q + O(h^2)$. No lower C^2 test exists there. At the origin the corresponding coefficient of $|h|^q$ is $K\lambda^{q/2} - A < 0$, with the same conclusion. Thus the supersolution condition, including its strict margin, is vacuous on this plane. For $q = 2$ the barrier is the smooth polynomial $mz + K|\xi|^2 - 3K\lambda z^2$, so the inequality already proved for $z \neq 0$ extends to $z = 0$ by continuity. Finally $B_v^- = -B_{-v}^+$ transfers all the lower-barrier assertions, including the viscosity tests on the plane. □

For the next two statements, assume $u \in C(\overline{\Omega} \cap \overline{B}_1)$ and

$$\begin{aligned} &\Omega \cap B_1 = \{x \in B_1 : x_n > \psi(x')\}, \qquad \psi, G \in C^{1,\gamma}(B_1'), \quad 0 < \gamma \leq 1, \\ &\psi(0) = D\psi(0) = G(0) = 0, \qquad DG(0) = p, \qquad [D\psi]_{C^{0,\gamma}(B_1')} + [DG]_{C^{0,\gamma}(B_1')} \leq \eta, \\ &u(x', \psi(x')) = G(x') \text{ on the true boundary}, \quad \|u - e\cdot x\|_{L^\infty(\Omega\cap B_1)} \leq \varepsilon, \quad e = (p, b_n). \end{aligned} \tag{4.12}$$

The full convex base disk is part of the hypotheses; only graph points in the comparison ball impose boundary values for that comparison.

Lemma 4.4 (Forced boundary sandwich)**.** *Let* (4.12) *hold, with* $|e| \leq 2$, *and let* $-F \leq -\Delta_\infty u \leq F$. *Put* $q = 1 + \gamma$. *There is* $c_q > 0$ *such that*

$$0 < \delta < \delta_0, \qquad \varepsilon + \eta \leq c_q\delta^{q+1}, \qquad F \leq c_q\delta^3 \tag{4.13}$$

imply a sandwich $B_- \leq u \leq B_+$ *in* $\Omega \cap B_1$, *with* $B_\pm(0) = 0$ *and* $|DB_\pm - e| \leq C_q\delta$.

Proof. Choose $s_\pm \in \{\delta, 2\delta\}$ so that $|b_n + s_+|, |b_n - s_-| \geq \delta/2$. Set $v_\pm = e \pm s_\pm e_n$ and use respectively $B_{v_+}^+$ and $B_{v_-}^-$. Since $\delta < \delta_0 \leq 1/2$, their lengths lie in $[\delta/2, 3]$. The parameters μ, k, δ_0 are those fixed in the preceding lemma. By decreasing the smallness constant in (4.13),

we may use $C\eta \le \delta/12$, since $\delta \le 1$ and $\eta \le c_q\delta^{q+1} \le c_q\delta$. On the graph, writing $r = |x'|$, Taylor's formula gives $|\psi(x')| + |G(x') - p\cdot x'| \le C\eta r^q$. For either vector $v = mw$,

$$|P_{w^\perp}x| \ge (|v_n|/m)r - |\psi(x')| \ge \frac{\delta}{4m}r, \qquad |z| \le |x| \le 2r.$$

Here $m \le 3$ and $C\eta \le \delta/12$ justify the projection bound. The fixed choice $\frac{4}{q-1}(8\mu)^q \le 1/4$ gives

$$KS^q - A|z|^q \ge \tfrac{3}{4}K(\delta/(4m))^q r^q \ge c_q\delta^{q+1}r^q.$$

The graph error $|v_\pm \cdot x - G(x')|$ is at most $4C\eta r^q$, since $|(v_\pm)_n| \le 3$. Decreasing the same smallness constant absorbs this error and proves the graph ordering.

On $\overline{\Omega \cap B_1} \cap \partial B_1$, put $t = x_n \ge -C\eta$ and $\alpha = |v_n|/m \ge \delta/(2m)$. The relevant gap for either sign is $st + KS^q - A|z|^q$. If $t \ge \alpha/2$, its tilt is at least $\delta^2/(4m)$, while

$$\frac{A}{\delta^2/(4m)} = \frac{16}{q-1}k\mu^{q+1}(\delta/m)^{q-1} \le \frac{16}{q-1}k\mu^{q+1}2^{q-1}.$$

The choice of k in the preceding lemma bounds this ratio by $1/4$. The remaining gap is therefore at least

$$\frac{3\delta^2}{16m} \ge \frac{\delta^2}{16} \ge \frac{\delta^{q+1}}{16}.$$

Otherwise $|t| \le \alpha/2$, by smallness of η. For unit vectors w, x, the decomposition $w = (w\cdot x)x + (w - (w\cdot x)x)$ and the identity $|w - (w\cdot x)x| = |P_{w^\perp}x|$ imply $|w_n| \le |x_n| + |P_{w^\perp}x|$. Thus $S \ge \alpha/2 \ge \delta/(4m)$. Since $|z| \le 1$,

$$\frac{A|z|^q}{KS^q} \le \frac{4}{q-1}(4\mu)^q \le \frac{1}{4}.$$

It follows that $KS^q - A|z|^q \ge (3/4)K(\delta/(4m))^q \ge c_q\delta^{q+1}$, while the negative tilt costs only $C\delta\eta$. Decrease the constant in (4.13) to absorb it and the spherical error ε. Finally decrease its forcing constant below half the strict PDE margin of Lemma 4.3; Lemma 2.1 proves the sandwich. The definitions give $B_\pm(0) = 0$, and $|DB_\pm - e| \le |DB_\pm - v_\pm| + s_\pm \le (C_q + 2)\delta$. □

4.4. **Recentered geometry and the boundary flatness estimate.**

Proposition 4.5 (Quantitative boundary flatness)**.** *Assume* (4.12), $|e| \le 1$, *and* $-F \le -\Delta_\infty u \le F$. *There are* $E_*, F_*, C > 0$, *depending only on* n, γ, b, C_I, *such that, if* $\varepsilon + \eta \le E_*$ *and* $F \le F_*$, *then*

$$\|Du - e\|_{L^\infty(\Omega\cap B_{1/512})} \le C\left((\varepsilon+\eta)^{1/(2+\gamma)} + F^{1/3}\right)^{b/(1+b)}. \tag{4.14}$$

Proof. Reduce E_* so that $\eta \le 1/100$. Apply Lemma 2.2 in the following fixed neighborhoods: $V = B'_{1/2} \times (-1/2, 1/2) \Subset B_1$, $U_1 = B_{1/8}$, and, for example, $U_0 = B_{1/16}$ in that lemma. The graph lies in this cylinder, since $|\psi(\xi)| \le \eta|\xi|^{1+\gamma}$, and $K = \|D\psi\|_\infty \le \eta$, $M = [D\psi]_\gamma \le \eta$ there. The buffer in (2.1) is $3/8$. Thus $\rho = r_b := 1/64$ satisfies both restrictions in (2.2). For every $z = X(\xi_0) \in \partial\Omega \cap B_{1/8}$, the lemma gives a tangent frame P_z, the inward normal ν_z, and $R_z = (P_z, \nu_z)$, with an exact upper-graph representation on $B_{r_b}(z)$ and a full tangent-coordinate disk B'_{2r_b}. Its estimates, and $\|DG\|_\infty \le |p| + \eta \le 2$, give

$$\psi_z(0) = D\psi_z(0) = 0, \qquad [D\psi_z]_\gamma + [DG_z]_\gamma \le C\eta.$$

Define the compatible slope

$$a_z = e + J_0(J_0^TJ_0)^{-1}(DG(\xi_0) - J_0^Te).$$

Here $J_0 = DX(\xi_0)$, and $DG(\xi_0) - J_0^Te = DG(\xi_0) - p - b_nD\psi(\xi_0)$. Since $\|J_0(J_0^TJ_0)^{-1}\| \le 1$, we obtain $|a_z - e| \le C\eta$ and $|a_z| \le 2$. Formula (2.9) gives $P_z^Ta_z = DG_z(0)$, so this slope is compatible with the trace. Set

$$\widetilde{u}_z(Y) = \frac{u(z + r_bR_zY) - g(z)}{r_b}.$$

Its affine slope is $R_z^T a_z$; its affine error is at most $2\varepsilon/r_b + C\eta$, the sum of the graph and trace derivative seminorms is at most $C\eta r_b^\gamma$, and its forcing bound is $r_b F$. These facts follow respectively by subtracting the two original affine errors, by the preceding derivative formulas, and by first-order scaling of the viscosity inequalities.

Put $E = \varepsilon + \eta$ and choose

$$\delta = C_0(E^{1/(2+\gamma)} + F^{1/3}).$$

Take C_0 large enough for the smallness conditions of Lemma 4.4, including all recentering constants, and then E_*, F_* small enough that $\delta < \delta_0 \le 1$. The case $E = F = 0$ is immediate. The sandwich at every such z gives, in the original coordinates,

$$|u(y) - g(z) - a_z \cdot (y - z)| \le C\delta|y - z|, \qquad y \in \overline{\Omega} \cap B_{r_b}(z). \tag{4.15}$$

Indeed, integrate the barrier gradient errors along the straight segment in the full ball on which the barriers are defined; that segment need not lie in Ω.

For $x \in \Omega \cap B_{1/512}$, set $d = d(x)$ and choose any nearest boundary point z. Since $0 \in \partial\Omega$, $d \le |x|$ and $|z| \le 2/512 < 1/8$. For all $y \in B_{d/4}(x)$, the same point z satisfies $|y - z| \le 5d/4 < r_b$. Thus (4.15) gives

$$\|u - A_z\|_{L^\infty(B_{d/4}(x))} \le C\delta d, \qquad A_z(y) = g(z) + a_z \cdot (y - z).$$

The normalized oscillation, trace bound, and $F \le 1$ give a uniform ambient Lipschitz bound on $\overline{\Omega} \cap B_{1/4}$ by Lemma 2.3. In particular $\operatorname{osc}_{B_{d/2}(x)} u \le Cd$. Applying (4.1) to this ball gives $[Du]_{C^{0,b}(B_{d/4}(x))} \le Cd^{-b}$. Directional difference quotients, with $h = (d/8)\delta^{1/(1+b)}$, therefore yield

$$|Du(x) - a_z| \le C\left(\frac{\delta d}{h} + d^{-b}h^b\right) \le C\delta^{b/(1+b)}.$$

Indeed, $h \le d/8$ because $\delta \le 1$. For each unit vector, subtract the affine fits at x and at its increment of length h; the two fitting errors contribute $C\delta d/h$. Integration of the gradient along that segment contributes at most $Cd^{-b}h^b$. Taking the supremum over unit vectors proves the displayed estimate. Since $|a_z - e| \le C\eta$ and $\eta \le C\delta^{2+\gamma}$, this proves (4.14). □

4.5. **The general-data boundary theorem.**

Theorem 4.6 (Boundary C^1 regularity and a uniform qualitative modulus)**.** *Let a boundary patch and its trace be $C^{1,\gamma}$, $0 < \gamma \le 1$. Every function continuous in the domain and up to that patch, and satisfying $-F \le -\Delta_\infty u \le F$, belongs to C^1 up to a smaller patch. Here C^1 includes the relative first-order expansion at boundary points and the continuous extension of the interior gradient.*

For families with common graph neighborhoods and positive smaller-patch buffers, uniformly bounded $C^{1,\gamma}$ graph and trace norms, solution oscillations, and forcing bounds, there is a common function $\omega(t) \downarrow 0$ as $t \downarrow 0$ such that

$$|Du(x) - Du(y)| \le \omega(|x - y|) \tag{4.16}$$

on the smaller closed patches. The modulus depends only on those bounds and on n, γ, b, C_I; it requires no common modulus of continuity of a specified source.

Proof. Proposition 4.2 gives a boundary slope p_z at every boundary point. Restricting its expansion to the graph gives $P_{T_z} p_z = D_T g(z)$, where P_{T_z} is tangent projection. The local Lipschitz bound gives $|p_z| \le L$. This follows also by testing its affine blowup in all inward directions; the supremum of the absolute value of a linear functional over a hemisphere is its full norm. Fix $M_0 \ge 2\max\{1, L\}$ and use tangent coordinates at z. For

$$u_r(Y) = \frac{u(z + rR_z Y) - g(z)}{M_0 r},$$

the boundary affine slope has norm at most $1/2$, the affine error tends to zero by (4.5), the graph and trace derivative seminorms are $O(r^\gamma)$, and the forcing is rF/M_0^3. Lemma 2.2

supplies tangent graphs and trace norms on common smaller disks. Proposition 4.5 therefore gives

$$\lim_{\substack{x\to z\\ x\in\Omega}} Du(x) = p_z.$$

To include variation along the boundary, if $z_j \to z$, choose interior points $x_j \to z$ with $|Du(x_j) - p_{z_j}| < 1/j$. The preceding limit gives $p_{z_j} \to p_z$. This proves the first assertion.

For uniformity, fix one $M_0 \geq 2\max\{1, L_{\text{family}}\}$, where L_{family} is the common ambient Lipschitz bound. Suppose the conclusion fails in a family with the stated fixed bounds. Restrict to a common buffered neighborhood and pass to a subsequence of the graph functions and coordinate traces converging in C^1. Their $C^{1,\gamma}$ bounds pass to the limit. Subtract the boundary value at graph points with one fixed parameter from each solution and its trace. Use the McShane Lipschitz extension to a common compact ambient neighborhood, inside the larger buffered chart. The extensions U_j have the same Lipschitz bound. Their anchors and the bounded size of the neighborhood give a common bound on their values, so Arzelà–Ascoli gives a uniformly convergent subsequence with limit U_∞. The restriction u_∞ to the limiting graph domain has the limiting trace and satisfies the same two-sided viscosity inequalities by stability: every compact subset of the limiting open graph domain lies in all sufficiently late domains. On interior compact subsets the gradients converge as well, by (4.1) and uniqueness of the derivative of the uniform limit.

There are points x_j, y_j in the smaller closed patches with $|x_j - y_j| \to 0$ and $|Du_j(x_j) - Du_j(y_j)| \geq \epsilon_0 > 0$. After a subsequence these points have a common limit. If it were interior, the preceding gradient convergence would contradict their separation. Write the boundary limit as $z_\infty = X_\infty(\xi_\infty)$ and choose the exact corresponding boundary centers $z_j = X_j(\xi_\infty)$. Let $p_\infty = Du_\infty(z_\infty)$ and define

$$a_j = p_\infty + D_T g_j(z_j) - P_{T_j} p_\infty,$$

where P_{T_j} denotes tangent projection at z_j. The tangential identity for the limiting boundary expansion gives $P_{T_\infty} p_\infty = D_T g_\infty(z_\infty)$. Together with C^1 convergence of the graphs and traces, it shows that a_j has the correct tangential component and $a_j \to p_\infty$; in particular $|a_j|/M_0 \leq 1$ for all sufficiently large j. The tangent frames constructed from the graph matrices converge too.

Fix a small radius r before taking $j \to \infty$. Put $h_j = \|\psi_j - \psi_\infty\|_\infty$ in the common coordinates. For $y \in \overline{\Omega}_j \cap B_r(z_j)$, the point

$$y^* = (y', \max\{y_n, \psi_\infty(y')\})$$

belongs to $\overline{\Omega}_\infty$, satisfies $|y - y^*| \leq h_j$, and lies in $B_{2r}(z_\infty)$ for all sufficiently large j. Uniform convergence of the extensions and their Lipschitz bound imply

$$\begin{aligned}
&\frac{1}{r} \sup_{\overline{\Omega}_j \cap B_r(z_j)} |u_j(y) - g_j(z_j) - a_j \cdot (y - z_j)| \\
&\quad \leq \frac{1}{r} \sup_{\overline{\Omega}_\infty \cap B_{2r}(z_\infty)} |u_\infty(y) - g_\infty(z_\infty) - p_\infty \cdot (y - z_\infty)| \\
&\qquad + \frac{C}{r} \left(\|U_j - U_\infty\|_\infty + h_j + |z_j - z_\infty|\right) + |a_j - p_\infty|.
\end{aligned}$$

Thus the normalized affine errors can first be made small by fixing r small, and then by taking j large. At the same time the uniform graph and trace seminorms scale as $O(r^\gamma)$ and the forcing as $O(r)$. Apply Proposition 4.5 in the tangent chart at z_j, with the common amplitude normalization M_0. More precisely, if $E_j(r)$ denotes the sum of the normalized affine error and the two rescaled derivative seminorms, the estimate gives

$$\sup_{\Omega_j \cap B_{r/512}(z_j)} |Du_j - a_j| \leq M_0 C \left(E_j(r)^{1/(2+\gamma)} + (rF/M_0^3)^{1/3} \right)^{b/(1+b)}.$$

The displayed affine-error bound and differentiability of u_∞ give $\lim_{r\downarrow 0}\limsup_{j\to\infty} E_j(r)=0$. Choose r first so that this right-hand side has upper limit less than $\epsilon_0/3$, and then take j large enough that the flatness hypotheses hold and the right-hand side is less than $\epsilon_0/3$. Thus all gradients in $B_{r/512}(z_j)$ are within $\epsilon_0/3$ of a_j. The estimate includes boundary points by the continuity already proved for each individual solution. Both x_j and y_j lie in this ball for large j, contradicting their fixed gradient separation. The supremum of all gradient differences at separation at most t in the bounded family is therefore a finite modulus tending to zero. □

This proves Theorem 1.2. It supplies a qualitative gradient modulus for general data. It does not assign a power rate to the boundary affine expansion. Quantitative boundary Hölder estimates require a further uniform affine-decay hypothesis or one of the structural arguments developed in the subsequent sections.

5. Flat constant boundary traces

Write $B_R^+=B_R\cap\{x_n>0\}$; its flat face is $B_R\cap\{x_n=0\}$.

Theorem 5.1 (Flat constant-trace boundary)**.** *Let $R>0$, $F\ge 0$, and $u\in C(\overline{B_R^+})$ satisfy $-F\le -\Delta_\infty u\le F$ in B_R^+ and $u=c$ on its flat face. Then $u\in C^{1,b_0}(\overline{B_{R/2}^+})$, and (3.7) holds with $B_{R/2}$ replaced by $\overline{B_{R/2}^+}$ and B_R by B_R^+, with a modified constant. If, for some $0<\sigma\le 1$, we also have $-\Delta_\infty u=f$ with $f\in C^{0,\sigma}(\overline{B_R^+})$ and $f=0$ on the flat face, then $u\in C^{1,b_\sigma}(\overline{B_{R/2}^+})$ and (3.8) holds with the same domain replacements. The constants have the dependence stated in Theorem 3.3.*

Proof. Set $v=u-c$, and define $U(x',t)=v(x',t)$ for $t\ge 0$ and $U(x',t)=-v(x',-t)$ for $t<0$. The zero trace makes $U\in C(\overline{B}_R)$. Reflection and multiplication by -1 preserve the two bounds on $-\Delta_\infty U$ away from the plane. At $z=(x_0',0)\in B_R$, normalize an upper test by $\phi(z)=U(z)=0$. Its tangential gradient vanishes because $U=0$ on the plane. For all sufficiently small h, oddness and upper contact give

$$0=U(z+he_n)+U(z-he_n)\le \phi(z+he_n)+\phi(z-he_n)=h^2\phi_{nn}(z)+o(h^2).$$

Thus $\phi_{nn}\ge 0$, and hence $-\Delta_\infty\phi=-(\phi_n)^2\phi_{nn}\le 0\le F$. Applying this upper-test calculation to $-U$ gives the lower-test inequality for U. The operator formula also covers $\phi_n=0$. Consequently the two bounded inequalities hold throughout B_R.

In the source case, define $f_o(x',t)=f(x',t)$ for $t>0$, $f_o(x',t)=-f(x',-t)$ for $t<0$, and $f_o=0$ on the plane. Reflection gives $-\Delta_\infty U=f_o$ off the plane, and the preceding interface test proves this equation on the plane as well. For points x,y on opposite sides, the projections $(x',0)$ and $(y',0)$ lie on the flat face. Its vanishing source therefore gives

$$|f_o(x)-f_o(y)|\le [f]_{\sigma;\overline{B_R^+}}(|x_n|^\sigma+|y_n|^\sigma)\le 2^{1-\sigma}[f]_{\sigma;\overline{B_R^+}}|x-y|^\sigma.$$

Here $|x_n|+|y_n|\le|x-y|$ and $s^\sigma+t^\sigma\le 2^{1-\sigma}(s+t)^\sigma$ for $s,t\ge 0$. On either half-ball reflection preserves the original Hölder bound. Hence $f_o\in C^{0,\sigma}(B_R)$, with

$$\|f_o\|_{L^\infty(B_R)}=\|f\|_{L^\infty(B_R^+)},\qquad [f_o]_{\sigma;B_R}\le 2^{1-\sigma}[f]_{\sigma;B_R^+}.$$

Apply Theorem 3.3 to U, using its bounded-inequality estimate in the first case and its Hölder-source estimate in the second. The trace value c lies between the limiting extrema of u, so

$$\operatorname{osc}_{B_R} U=2\|u-c\|_{L^\infty(B_R^+)}\le 2\operatorname{osc}_{B_R^+} u.$$

The same interior theorem, applied on balls compactly contained in B_R, also gives $U\in C^1_{\mathrm{loc}}(B_R)$. Its gradient bounds on $B_{R/2}$ therefore extend by continuity to $\overline{B}_{R/2}\Subset B_R$. Restricting $U+c=u$ to the closed upper half-ball gives the relative boundary expansion and the full pairwise gradient estimate, including the spherical rim. The oscillation and reflected-source bounds yield the stated half-ball estimates. Without the boundary vanishing condition, the reflected source need not be continuous; the first assertion still applies. □

6. Homogeneous replacement and constant boundary traces

We retain the pointwise viscosity convention of Section 2 and the constants a, C_H fixed in (1.3). All distances and diameters are Euclidean unless a path distance is displayed. The homogeneous estimate (1.2), scaled to a full ball of radius s, gives

$$\|DH\|_{L^\infty(B_{s/2})} + s^a [DH]_{C^{0,a}(B_{s/2})} \le \frac{C_H}{s} \operatorname{osc}_{B_s} H. \tag{6.1}$$

This applies to the n-dimensional replacements by the dimension extension already established in Section 1.

6.1. The gradient-patching input and its error.

The following is the upper version of the Jensen patching theorem. Its viscosity formulation at the interface of the patched set is essential here.

Theorem 6.1 (Jensen patching, [8]). *Let $D \subset \mathbb{R}^n$ be a bounded domain and let $h \in C(\overline{D})$ be infinity harmonic in D. For every $t > 0$, there is $h_t^+ \in C(\overline{D})$ with $h_t^+ = h$ on ∂D, such that every lower C^2 test θ for h_t^+ at any point of D satisfies*

$$-\Delta_\infty \theta \ge 0, \qquad |D\theta| \ge t. \tag{6.2}$$

To describe the function, put

$$V_t = \{x \in D : |Dh(x)| < t\}.$$

Then $h_t^+ = h$ on $D \setminus V_t$. On each connected component U of V_t,

$$h_t^+(x) = \inf_{y \in \partial U} \{h(y) + t\, d_U(x, y)\}, \qquad x \in U. \tag{6.3}$$

Here d_U is the infimum of lengths of curves whose interiors lie in U and whose endpoints are the indicated points. A boundary point not accessible by a curve of finite length contributes $+\infty$ to the infimum.

The cited statement follows by applying [8, Lemma 2.2 and Theorem 2.3] to $-h$ and changing signs. Lemma 2.2 supplies the patch construction; Theorem 2.3 identifies it with the solution of Jensen's gradient-constraint equation and therefore gives (6.2) at every interface point as well as inside each component. More explicitly, a lower test θ for h_t^+ corresponds to the upper test $-\theta$ for the lower patch of $-h$. The upper-test inequalities in [8, Theorem 2.3] are $\Delta_\infty(-\theta) \ge 0$ and $|D(-\theta)| \ge t$, which are exactly (6.2). These results require a bounded domain, with no convexity or boundary smoothness hypothesis. For a bounded open set that is not connected, apply the construction on each component. The Euclidean error bound proved below ensures continuity at boundary accumulation points of those components. In the present setting Dh is continuous by (6.1), so V_t is open.

Lemma 6.2 (Euclidean patching error). *The function in Theorem 6.1 satisfies*

$$0 \le h_t^+(x) - h(x) \le 2t \operatorname{dist}(x, \partial D) \le 2t \operatorname{diam} D, \qquad x \in D. \tag{6.4}$$

In particular, no intrinsic-diameter bound is needed for this estimate.

Proof. There is nothing to prove outside V_t. Fix a component U of V_t and $x \in U$. Along any curve of finite length whose interior lies in U, integration of $|Dh| < t$, followed by continuity at an endpoint in ∂U, gives

$$h(x) - h(y) \le t\, d_U(x, y).$$

Formula (6.3) therefore gives $h_t^+(x) \ge h(x)$.

Because U is bounded and open, a nearest point $y_* \in \partial U$ exists. Put $d_* = \operatorname{dist}(x, \partial U) > 0$. The segment from x to y_*, excluding y_*, lies in $B_{d_*}(x) \subset U$. Thus $d_U(x, y_*) = |x - y_*| = d_*$ and $|h(x) - h(y_*)| \le t d_*$. Taking this particular endpoint in (6.3) gives

$$h_t^+(x) - h(x) \le 2t d_*.$$

Since $U \subset D$, $\operatorname{dist}(x, \partial U) \le \operatorname{dist}(x, \partial D)$. This proves (6.4). It also shows directly that the patch tends to the same boundary values as h. □

6.2. **A quantitative replacement on an arbitrary bounded set.**

Lemma 6.3 (Homogeneous replacement with bounded forcing)**.** *Let $D \subset \mathbb{R}^n$ be nonempty, bounded, and open, let $u \in C(\overline{D})$ satisfy*

$$-F \le -\Delta_\infty u \le F \quad \text{in } D, \qquad F \ge 0, \tag{6.5}$$

and let $h \in C(\overline{D})$ be infinity harmonic with $h = u$ on ∂D. Set

$$M = \operatorname{osc}_D u, \qquad R = \operatorname{diam} D.$$

Then

$$\|u - h\|_{L^\infty(D)} \le 32\, M^{2/5} R^{4/5} F^{1/5}. \tag{6.6}$$

Proof. If $M = 0$, then u and h are the same constant. If $F = 0$, homogeneous comparison gives $u = h$. We may therefore suppose $M, F > 0$; in particular $R > 0$. Fix $x_0 \in D$. Under the change of variables

$$x = x_0 + Ry, \qquad \widetilde{u}(y) = \frac{u(x_0 + Ry) - \inf_D u}{M},$$

with h rescaled by the same translation and amplitude, the domain has diameter one, the oscillation of $\widetilde{u}$ is one, and the right-hand-side bound is $\widetilde{F} = R^4 F / M^3$. The homogeneous maximum principle gives $0 \le \widetilde{h} \le 1$, and by construction $0 \le \widetilde{u} \le 1$ as well. We drop the tildes.

First assume $0 < F \le 32^{-5}$ and set

$$t = F^{1/5}, \qquad \eta = 16t \le \frac{1}{2}.$$

Theorem 6.1 and Lemma 6.2 give a function $v = h_t^+$ such that

$$h \le v \le h + 2t, \qquad v = h = u \quad \text{on } \partial D,$$

and (6.2) holds for every lower test of v. In particular, $0 \le v \le 1 + 2t < 2$. For $|s| \le 2$, define

$$\phi(s) = s + \frac{\eta}{4}(4 - s^2), \qquad w = \phi(v).$$

The function ϕ is strictly increasing near the range of v, with

$$\phi' \ge \frac{1}{2}, \qquad \phi'' = -\frac{\eta}{2}, \qquad 0 \le \phi(v) - v \le \eta.$$

To verify the PDE inequality for w without assuming smoothness of v, let ψ be a lower test for w. At its contact point x, we have $v(x) \in [0, 1 + 2t] \Subset (-2, 2)$. Since $\phi' > 0$ on $(-2, 2)$, its inverse is smooth on $\phi((-2, 2))$. Restrict the contact neighborhood so that ψ takes values there. Monotonicity then shows that $\theta = \phi^{-1} \circ \psi$ is a lower test for v. Thus (6.2) gives $-\Delta_\infty \theta \ge 0$ and $|p| \ge t$, where $p = D\theta$ at contact. Applying the ordinary chain rule to these smooth tests gives

$$-\Delta_\infty \psi = (\phi')^3(-\Delta_\infty \theta) - (\phi')^2 \phi'' |p|^4 \ge \frac{\eta t^4}{8} = 2F.$$

Thus $-\Delta_\infty w \ge 2F$ in the viscosity sense. Also $w \ge u$ on ∂D. Applying Lemma 2.1 to u and w, with constants $F < 2F$, yields $u \le w$. It follows that

$$u - h \le (v - h) + (w - v) \le 2t + \eta = 18F^{1/5}.$$

Apply this upper estimate to $1 - u$ and its homogeneous replacement $1 - h$, both with values in $[0, 1]$, to obtain the lower bound.

If $F > 32^{-5}$, the normalized ranges of u and h give $\|u - h\|_\infty \le 1 \le 32F^{1/5}$. The normalized estimate is therefore valid for all $F > 0$. Reversing the scaling gives

$$32M \left(\frac{R^4 F}{M^3}\right)^{1/5} = 32 M^{2/5} R^{4/5} F^{1/5},$$

as asserted. □

Remark 6.4. The only comparison involving nonzero right-hand sides used here is the strict comparison of Lemma 2.1, so no comparison theorem for a prescribed sign-changing source is needed.

6.3. **The interior affine estimate needed below.**

Lemma 6.5 (Two-scale interior affine approximation)**.** *Suppose u satisfies (6.5) in $B_r(x)$, is continuous on its closure, and has Lipschitz constant at most L there. Put $A = L + F^{1/3}$. If $A > 0$, $0 < r \le 1$, and $0 < \rho \le r/4$, there is $p_{x,r} \in \mathbb{R}^n$, with $|p_{x,r}| \le CA$, such that*

$$\sup_{B_\rho(x)} |u(y) - u(x) - p_{x,r} \cdot (y - x)| \le CA(r^2 + \rho^{1+a} r^{-a}). \tag{6.7}$$

The constant depends only on the homogeneous interior constants in dimension $n + 1$.

Proof. Set $v(y) = (u(x + ry) - u(x))/(Ar)$ and $\delta = rF/A^3 \le r$. Then $v(0) = 0$, $|v| \le 1$, and $-\delta \le -\Delta_\infty v \le \delta$. Lemma 3.1, with $c = 0$, gives a homogeneous H in B_1^{n+1} satisfying $\|H - (v + 2t)\|_\infty \le \delta$. Homogeneous comparison with constants gives $\operatorname{osc} H \le \operatorname{osc}_{\partial B_1}(v + 2t) \le 6$. Theorem 1.1 and the anchored Taylor calculation in Proposition 3.2 give

$$|v(y) - D_y H(0, 0) \cdot y| \le 2\delta + C|y|^{1+a}, \qquad |y| \le 1/4.$$

Multiplication by Ar, and $Ar\delta = (F/A^2)r^2 \le Ar^2$, prove the estimate with $p_{x,r} = AD_y H(0, 0)$. The slope is bounded by CA. □

6.4. **Odd full-ball replacements near a curved zero trace.** Work in a boundary neighborhood slightly larger than the final patch. All charts, balls, and line segments in the construction below are contained in this larger neighborhood. Suppose that u has an ambient Lipschitz bound L there and that $u = 0$ on the physical boundary patch. For a $C^{1,\gamma}$ patch, Lemma 2.2 supplies tangent charts with a common positive radius and a common constant K so that, at each boundary center z, rigid coordinates centered at z give, within the coordinate cylinder,

$$\Omega = \{(x', t) : t > \psi_z(x')\}, \qquad \psi_z(0) = 0, \quad D\psi_z(0) = 0, \quad |\psi_z(x')| \le K|x'|^{1+\gamma}. \tag{6.8}$$

The graph functions in this assertion are defined on full base disks larger than every disk used below; the side description holds in full cylinders. Shrinking the common radius preserves fixed positive distances to the edges of the larger coordinate neighborhood.

Lemma 6.6 (Curved constant-trace replacement)**.** *Under the preceding geometric and Lipschitz conditions, suppose additionally that $-F \le -\Delta_\infty u \le F$ in the larger neighborhood. There are a uniform $s_0 > 0$ and infinity-harmonic functions $H_{z,s}$ in $B_s(z)$, for $0 < s < s_0$, such that*

$$\|u - H_{z,s}\|_{L^\infty(\Omega \cap B_s(z))} \le CLs^{1+\gamma} + CL^{2/5} F^{1/5} s^{6/5}, \tag{6.9}$$

$$\|DH_{z,s}\|_{L^\infty(B_{s/2}(z))} + s^a [DH_{z,s}]_{C^{0,a}(B_{s/2}(z))} \le CL. \tag{6.10}$$

The constants depend only on the fixed chart bounds and the homogeneous interior constants.

Proof. Choose the common radius s_0 sufficiently small that all balls $B_{4s}(z)$, $s < s_0$, lie in the larger Lipschitz neighborhood. Also require $Ks_0^\gamma \le 1$ and the validity of (6.8) on the corresponding cylinders and full base disks.

In these balls extend u by zero to the other side of the physical boundary, and call the extension U_0. It is Lipschitz with constant L. Indeed, for two points on different sides, their segment lies in the larger convex ambient ball and has a first boundary crossing w. Since $u(w) = 0$, the difference of the extended values is at most L times the distance to w, hence at most L times the distance between the original points. For two interior points use the assumed ambient Lipschitz bound; for two exterior points the difference is zero.

Set $z = 0$ in tangent coordinates. On the upper hemisphere of ∂B_s prescribe

$$\Phi(x', t) = U_0(x', t) - U_0(x', 0), \qquad t \ge 0,$$

and extend these values oddly to the lower hemisphere. The values are continuous at the equator, since they vanish there. Moreover $|\Phi(x',t)| \le L|t|$. Let H_s be the unique homogeneous solution in B_s with boundary values Φ. Reflection and sign invariance, followed by uniqueness, imply that H_s is odd in t. In the upper half-ball compare H_s with the affine functions Lt and $-Lt$; they order the values on both the hemisphere and the flat face. Oddness therefore gives

$$|H_s(x',t)| \le L|t| \quad \text{throughout } B_s. \tag{6.11}$$

We now check the boundary error on $D_s = \Omega \cap B_s$ in all possible locations. On the physical graph, $u = 0$ and (6.11) gives

$$|u - H_s| \le L|\psi_z(x')| \le LKs^{1+\gamma}.$$

On the upper spherical portion belonging to $\overline{\Omega}$, the prescribed boundary value of H_s gives $u - H_s = U_0(x',0)$. The point $(x',\psi_z(x'))$ lies in B_{2s} and has zero extended value. Thus

$$|U_0(x',0)| \le L|\psi_z(x')| \le LKs^{1+\gamma}.$$

The fact that this graph point can lie outside B_s is harmless: its norm is at most $\sqrt{s^2 + K^2 s^{2+2\gamma}} \le \sqrt{2}\, s < 2s$, and U_0 was defined and controlled on the fixed larger ball.

Finally, consider a lower spherical point of $\overline{\Omega}$. Then $\psi_z(x') \le t < 0$, so $|t| \le |\psi_z(x')|$. Comparison of $u(x',t)$ with its zero graph value gives

$$|u(x',t)| \le L(t - \psi_z(x')) \le L|\psi_z(x')|.$$

Together with (6.11), this gives $|u - H_s| \le 2LKs^{1+\gamma}$ on that portion. The equatorial and intersection points follow either from these same non-strict inequalities or by continuity. These cases exhaust ∂D_s.

Let h_s be the homogeneous extension of the continuous (in fact L-Lipschitz) boundary values $u|_{\partial D_s}$. The homogeneous Dirichlet result of Section 2 and comparison apply on this bounded open set, componentwise if necessary. The just-proved boundary bounds imply

$$\|h_s - H_s\|_{L^\infty(D_s)} \le 2LKs^{1+\gamma}.$$

Since $u(0) = 0$ and the ambient Lipschitz bound is L, we have $\operatorname{osc}_{D_s} u \le 2Ls$ and $\operatorname{diam} D_s \le 2s$. Lemma 6.3 therefore gives

$$\|u - h_s\|_{L^\infty(D_s)} \le CL^{2/5}F^{1/5}s^{6/5}.$$

The triangle inequality proves (6.9); the estimate also holds on the closure of D_s by continuity. Finally, (6.11) gives $\operatorname{osc}_{B_s} H_s \le 2Ls$, so (6.1) proves (6.10). The forcing bound was used only in the homogeneous replacement on D_s. Consequently, for each fixed pair (z,s), (6.9) also holds with F replaced by any $F_{z,s} \ge 0$ for which $-F_{z,s} \le -\Delta_\infty u \le F_{z,s}$ in D_s. The Lipschitz bound L is still taken on the larger neighborhood. □

6.5. **The curved-boundary Hölder estimate.**

Theorem 6.7 (Curved constant traces with bounded forcing)**.** *Let Ω have a $C^{1,\gamma}$ boundary patch, $0 < \gamma \le 1$, and let u be continuous up to that patch, with locally constant trace. Suppose that $-F \le -\Delta_\infty u \le F$ nearby and that the local ambient Lipschitz estimate on a fixed larger neighborhood is L. Set*

$$\lambda = \min\{\gamma, 1/5\}, \qquad \beta = \frac{a\lambda}{1 + a + \lambda}. \tag{6.12}$$

Then u belongs to $C^{1,\beta}$ on every fixed smaller closed patch E'. In normalized coordinates,

$$\|Du\|_{L^\infty(E')} + [Du]_{\beta;E'} \le C(L + F^{1/3}),$$

where C depends on the fixed chart bounds, the separation from the edges of the larger patch, and the homogeneous interior constants.

Proof. Work first on a connected graph chart, where the locally constant trace has a single value. The finite-cover convention of Section 2 combines these local conclusions on a compact patch. Subtract that trace value and put $A = L + F^{1/3}$. If $A = 0$, the assertion is immediate. Fix nested ambient neighborhoods $U_0 \Subset U_1$ of the final and larger patches, and let $\sigma = \operatorname{dist}(\overline{U}_0, \mathbb{R}^n \setminus U_1) > 0$. All Lipschitz bounds are taken on a still larger neighborhood containing the balls needed to construct the replacements. Let s_0 be the common replacement radius.

For a target radius ρ, set

$$r = \rho^{(1+a)/(1+a+\lambda)}.$$

Choose a fixed upper threshold for ρ so small that

$$8\rho \le r, \qquad r < \min\{1, s_0/8, \sigma/64\},$$

and all balls $B_{32r}(z)$ arising from boundary centers within distance r of U_0 stay in the larger Lipschitz neighborhood. These are restrictions on the fixed threshold, independent of x and ρ.

Let $x \in \overline{\Omega} \cap U_0$. If x is interior and $d(x) = \operatorname{dist}(x, \partial\Omega) \ge r$, apply Lemma 6.5 in $B_r(x)$. This includes $d(x) = r$: the open ball lies in Ω, and continuity and the ambient Lipschitz bound hold on its closure. It gives an affine function anchored at $u(x)$ whose error on $B_\rho(x)$ is at most

$$CA(r^2 + \rho^{1+a} r^{-a}) \le CA(r^{1+\lambda} + \rho^{1+a} r^{-a}).$$

If $d(x) < r$, choose any nearest boundary point z; when x is on the boundary, take $z = x$. The buffer restrictions imply that z belongs to the prescribed boundary patch. No uniqueness of the nearest point is required. Use $H = H_{z,8r}$ from Lemma 6.6. Both x and $\overline{\Omega} \cap B_\rho(x)$ lie in $B_{2r}(z)$. Thus every segment from x to a point of this clipped ball lies in $B_{4r}(z)$, where the derivative estimate for H is valid. Use

$$P(y) = u(x) + DH(x) \cdot (y - x).$$

The replacement errors at x and y, and the Taylor remainder of H along their segment, give

$$\sup_{\overline{\Omega} \cap B_\rho(x)} |u - P| \le CLr^{1+\gamma} + CL^{2/5} F^{1/5} r^{6/5} + CL\rho^{1+a} r^{-a}.$$

Since $L \le A$, $F \le A^3$, and $\lambda \le \gamma, 1/5$, this is at most

$$CA(r^{1+\lambda} + \rho^{1+a} r^{-a}).$$

This argument includes boundary centers.

For the chosen r, the two terms have the common order $\rho^{1+\beta}$, since

$$r^{1+\lambda} = \rho^{1+a} r^{-a} = \rho^{1+a\lambda/(1+a+\lambda)}.$$

We have obtained the required affine approximation on every clipped ball. The slopes are uniformly bounded by CA: this is part of Lemma 6.5 in the interior case, and follows from (6.10) at x in the boundary case. Thus both the error and the slope hypotheses of the affine criterion hold uniformly. Lemma 2.2(ii) supplies the common thickness constant for $E = \overline{\Omega} \cap U_1$ and centers in the smaller closed patch, after reducing the uniform radius threshold below the fixed buffers. Lemma 2.4 therefore applies to the anchored affine functions just constructed, and gives both $\|Du\|_\infty \le CA$ and $[Du]_\beta \le CA$ on that patch. □

The Lipschitz bound used here is furnished by Lemma 2.3; it is not an additional hypothesis beyond the bounded forcing, Lipschitz trace, and fixed chart bounds.

Corollary 6.8 (A source vanishing at the boundary)**.** *Suppose, in the setting of Theorem 6.7, that u solves $-\Delta_\infty u = f$ and, for some $\kappa_0, C_f \ge 0$ and a common $s_f > 0$,*

$$\sup_{\Omega \cap B_s(z)} |f| \le C_f s^{\kappa_0} \qquad (0 < s < s_f)$$

uniformly for the boundary centers, with all these balls contained in the larger chart neighborhood. Then the theorem holds with

$$\lambda = \min\{\gamma, (1+\kappa_0)/5\}, \qquad \beta = \frac{a\lambda}{1+a+\lambda}.$$

The normalized gradient norm and Hölder seminorm are bounded by $C(L + F^{1/3} + C_f^{1/3})$, *with the constant allowed to depend additionally on* κ_0 *and* s_f. *In particular, if* $f \in C^{0,\sigma}$, $0 < \sigma \le 1$, *and* $f = 0$ *on the patch, one may take* $\kappa_0 = \sigma$.

Proof. Put $A_f = L + F^{1/3} + C_f^{1/3}$; the case $A_f = 0$ is immediate. Shrink the common replacement radius to at most s_f. Use the localized form of (6.9) just proved, with $F_{z,s} = C_f s^{\kappa_0}$. The forcing error becomes $CL^{2/5}C_f^{1/5}s^{1+(1+\kappa_0)/5}$. In the boundary case of the preceding theorem take $s = 8r < s_f$. Since $L^{2/5}C_f^{1/5} \le A_f$ and $\lambda \le (1+\kappa_0)/5$, this error is at most $C_{\kappa_0}A_f r^{1+\lambda}$ for $r \le 1$. The geometric error has the same bound because $\lambda \le \gamma$. In the interior case, Lemma 6.5 gives $CA_f(r^2 + \rho^{1+a}r^{-a})$, and $r^2 \le r^{1+\lambda}$ because $\lambda \le \gamma \le 1$. Thus the same affine approximations have error $C_{\kappa_0}A_f(r^{1+\lambda} + \rho^{1+a}r^{-a})$ and slopes bounded by CA_f. The balancing and affine-criterion argument of Theorem 6.7 now applies with this λ. The radius threshold is also restricted by $8r < s_f$, giving the stated dependence on s_f. For a Hölder source vanishing on the patch, $|f(x)| \le [f]_{C^{0,\sigma}}|x-z|^\sigma$ gives the displayed hypothesis. □

Proof of Theorem 1.3. The flat assertions are Theorem 5.1. The curved assertions follow from Theorem 6.7 and Corollary 6.8, with the ambient Lipschitz bound from Lemma 2.3 on a larger patch. For the quantitative scaling, let $V(y) = u(z + Ry) - c$. It satisfies viscosity bounds $\pm R^4 F$, has gradient $DV = RDu$, and has zero trace. On the unit chart, the Lipschitz bound is at most $C(\|V\|_\infty + (R^4F)^{1/3})$. The normalized theorems and $[DV]_\beta = R^{1+\beta}[Du]_\beta$ prove (1.7). If the source is Hölder and vanishes on the patch, its rescaled seminorm is $R^{4+\sigma}[f]_\sigma$. On the bounded unit chart, its sup norm is also at most a chart-dependent constant times this seminorm: compare each source value with a boundary point where it vanishes. This gives the additional term in both the flat and curved estimates. All chart and buffer constants are those of the rescaled geometry. □

Remark 6.9 (The homogeneous specialization). When $F = 0$, the forcing error in (6.9) vanishes. The same proof then uses $\lambda = \gamma$ and gives the exponent $a\gamma/(1+a+\gamma)$ on curved constant-trace patches. On a flat patch the odd extension is itself infinity harmonic, so Theorem 1.1 gives exponent a directly. These are the corresponding homogeneous boundary estimates.

7. A direct transfer from boundary affine decay

The next argument avoids using the already reduced exponent b_0 in a second transfer. All balls and nearest boundary points below are taken in fixed, successively smaller neighborhoods of a C^1 graph patch. We use the thickness assertion of Lemma 2.2(ii), which requires only a Lipschitz graph and has a uniform radius threshold on a smaller patch.

Theorem 7.1 (Boundary affine decay with bounded forcing)**.** *Suppose* u *is locally Lipschitz for the ambient Euclidean distance up to a* C^1 *boundary patch, with bound* L, *and satisfies* $-F \le -\Delta_\infty u \le F$ *in its interior. Suppose also that for each boundary point* z *on a larger patch there is* p_z *such that, uniformly for* $0 < s < s_0$,

$$\sup_{\overline{\Omega}\cap B_s(z)} |u(y) - u(z) - p_z \cdot (y-z)| \le Ms^{1+\lambda}, \qquad 0 < \lambda \le 1. \tag{7.1}$$

Then, on a smaller closed patch,

$$u \in C^{1,\beta}, \qquad \beta = \frac{a\lambda}{1+a+\lambda}.$$

Its gradient bound and Hölder seminorm depend only on $n, a, C_H, \lambda, L, M, F$, *the graph thickness and the fixed radii and neighborhood separations.*

Proof. First $|p_z| \le L$. For a unit vector e with $e \cdot \nu(z) > 0$, the C^1 graph description puts $z + te$ in Ω for all sufficiently small $t > 0$. Apply (7.1) at radius $2t$ and use the ambient Lipschitz bound to obtain

$$|p_z \cdot e| \le L + 2^{1+\lambda} M t^{\lambda}.$$

Letting $t \downarrow 0$ gives $|p_z \cdot e| \le L$ on the open inward half-sphere. By continuity the same holds on its closure; changing e to $-e$ covers the remaining directions. Thus $|p_z| \le L$, uniformly in z.

Set $A = L + F^{1/3}$, with the case $A = 0$ immediate. Work at radii below a common threshold as follows. Choose nested graph neighborhoods $U_0 \Subset U_1$ as in Section 2, with the smaller closed patch $E' \subset \overline{\Omega} \cap U_0$ and all hypotheses valid on U_1. Put $E = \overline{\Omega} \cap U_1$. Choose $0 < r_1 \le 1$ such that

$$6r_1 < \operatorname{dist}(\overline{U}_0, \mathbb{R}^n \setminus U_1), \qquad 3r_1 < s_0.$$

Fix $x \in E'$, $0 < r < r_1$, and $0 < \rho \le r/4$. If $d(x) = \operatorname{dist}(x, \partial\Omega) \ge 2r$, then $\overline{B_r(x)} \subset \Omega \cap U_1$. Lemma 6.5 applies with the given Lipschitz and forcing bounds. It gives an affine function P with $P(x) = u(x)$, $|DP| \le CA$, and

$$\sup_{B_\rho(x)} |u - P| \le CA(r^2 + \rho^{1+a} r^{-a}), \qquad 0 < \rho \le r/4. \tag{7.2}$$

If $d(x) < 2r$, take any nearest boundary point z; for a boundary center take $z = x$. The buffer restriction puts z and $B_{3r}(z)$ inside U_1. Both x and all points of $E \cap B_\rho(x)$ lie in $B_{3r}(z)$. Apply (7.1) at radius $3r < s_0$ and anchor at x: subtract the boundary affine remainder at x from the one at y. The affine function $P(y) = u(x) + p_z \cdot (y - x)$ satisfies

$$\sup_{\overline{\Omega} \cap B_\rho(x)} |u - P| \le 2M(3r)^{1+\lambda}, \qquad |DP| \le L. \tag{7.3}$$

This case includes centers on the physical boundary.

For each sufficiently small ρ, choose $r = \rho^{(1+a)/(1+a+\lambda)}$. The radius is below every fixed buffer, $r < r_1$, and $\rho \le r/4$ once a single uniform threshold for ρ is fixed. Since $\lambda \le 1$ and $r \le 1$, $r^2 \le r^{1+\lambda}$. Moreover,

$$r^{1+\lambda} = \rho^{1+a} r^{-a} = \rho^{1+\beta}.$$

Thus every center of the smaller closed patch has uniformly bounded anchored affine approximations with error $C(A + M)\rho^{1+\beta}$ on its entire clipped ball.

After reducing this uniform threshold if necessary, Lemma 2.2(ii) gives the thickness of $E \cap B_\rho(x)$ for every $x \in E'$. Since $0 < \beta < 1$, Lemma 2.4 applies with slope bound CA and error coefficient $C(A+M)$. It yields a gradient field with norm at most CA and a full pairwise $C^{0,\beta}$ estimate on E', including boundary centers. At a boundary point this relative gradient agrees with p_z by uniqueness of the first-order expansion in that lemma. Finally, $u(z) = g(z)$ identifies (1.8) with (7.1), so this also proves Theorem 1.4. □

8. Calibrated slopes and the range of possible exponents

8.1. **Forcing changes a calibrated boundary slope.** The exact global Lipschitz bound used for homogeneous calibrated chords does not persist for a forced equation. The following example shows explicitly why the prescribed boundary vector cannot be carried over unchanged.

Proposition 8.1 (A smooth calibration example)**.** *There are a smooth uniformly convex domain, smooth Dirichlet data, and a smooth positive source for which the homogeneous transverse chord calibration holds on a boundary patch, while the boundary derivative differs from its homogeneous calibrated vector.*

Proof. Let $\Omega = B_1 \subset \mathbb{R}^n$ and $0 < \varepsilon < 1/2$. Set

$$u(x) = x_n + \varepsilon(1 - |x|^2).$$

Then $u = g = x_n$ on ∂B_1, $Du = e_n - 2\varepsilon x$, and $D^2u = -2\varepsilon I$. Consequently

$$-\Delta_\infty u = 2\varepsilon|e_n - 2\varepsilon x|^2 =: f(x) > 0$$

on the closed ball. Indeed, $|e_n - 2\varepsilon x| \geq 1 - 2\varepsilon$, so $f \geq 2\varepsilon(1-2\varepsilon)^2 > 0$ there. Thus u is a classical, hence viscosity, solution with smooth source.

The Euclidean Lipschitz constant of g on the sphere is one: the coordinate projection is 1-Lipschitz, with equality at e_n and $-e_n$. The affine function $h(x) = x_n$ is infinity harmonic and has the same boundary values, so homogeneous uniqueness identifies it as the homogeneous Dirichlet solution. In particular, $Dh = e_n$. Fix $0 < \tau < 1$ and a boundary point z with $z_n \leq -\tau$. Put

$$b_z = z - 2z_n e_n, \qquad e_z = e_n.$$

Reflection of the last coordinate shows that $b_z \in \partial B_1$ and $(b_z - z)/|b_z - z| = e_z$. Moreover

$$g(b_z) - g(z) = -2z_n = |b_z - z|, \qquad |b_z - z| \geq 2\tau.$$

The inward unit normal is $\nu(z) = -z$, so $e_z \cdot \nu(z) = -z_n \geq \tau$. Thus the transverse chord calibration holds uniformly on the south cap with $L = 1$.

However,

$$Du(z) = e_n - 2\varepsilon z \neq e_n = Le_z.$$

In fact, the normal derivative differs by 2ε: $(Du(z) - e_n) \cdot \nu(z) = 2\varepsilon$. The global Lipschitz bound also changes: convexity of $\overline{B}_1$ and $|Du| \leq 1 + 2\varepsilon$ give $\mathrm{Lip}_{\overline{B}_1} u \leq 1 + 2\varepsilon$. Inward difference quotients at $-e_n$, where $Du(-e_n) = (1+2\varepsilon)e_n$, give the reverse inequality. Hence $\mathrm{Lip}_{\overline{B}_1} u = 1 + 2\varepsilon > 1$. This proves the assertion and the failure of the homogeneous Lipschitz bound. □

The gradient in this example is smooth, so the failure of the calibrated formula is compatible with boundary Hölder regularity.

8.2. **The universal upper obstruction.**

Proposition 8.2 (A smooth source and the exponent 1/3)**.** *Even on a flat constant-trace patch, a universal boundary gradient Hölder exponent cannot exceed* 1/3*.*

Proof. In the upper half-space set

$$u(x', t) = -\frac{3^{4/3}}{4} t^{4/3}, \qquad t \geq 0.$$

For $t > 0$, this is a classical solution with

$$u_t = -3^{1/3} t^{1/3}, \qquad u_{tt} = -3^{-2/3} t^{-2/3}, \qquad -\Delta_\infty u = -(u_t)^2 u_{tt} = 1.$$

Thus it is also a viscosity solution in the open half-space, with the smooth source $f \equiv 1$. Its trace on $t = 0$ is zero. For each boundary point $z = (z', 0)$,

$$|u(x', t) - u(z)| \leq \frac{3^{4/3}}{4} |(x' - z', t)|^{4/3},$$

which proves that its relative boundary gradient is zero. The interior gradient $Du = -3^{1/3} t^{1/3} e_n$ extends continuously to that boundary value. For $\beta > 1/3$ the quotient $|Du(0,t) - Du(0,0)|/t^\beta = 3^{1/3} t^{1/3-\beta}$ diverges as $t \downarrow 0$. Restricting to $B_1 \cap \{t > 0\}$ and taking the smaller flat patch $\{(x', 0) : |x'| < 1/2\}$ keeps the patch away from the spherical rim. The boundary geometry and trace are smooth on this patch, so the local assertion follows. □

8.3. **The additional estimate for general nonconstant data.** For a general $C^{1,\gamma}$ trace on a $C^{1,\gamma}$ boundary patch and bounded forcing, Theorem 1.2 gives boundary C^1 on smaller closed patches with a uniform qualitative modulus under its stated bounds. To obtain $C^{1,\beta}$ from the present method it is sufficient to establish the uniform power decay (1.8); the ambient Lipschitz bound from Lemma 2.3 and Theorem 1.4 then give its precise consequence. Pointwise differentiability alone does not supply that decay. The forced chord example shows why the homogeneous prescribed normal slope cannot be used without a new source-dependent argument. This is a limitation of the proved extension, not a counterexample to general boundary Hölder regularity. The unrestricted question already contains the corresponding homogeneous problem as the case $f = 0$.

References

[1] R. Jensen, *Uniqueness of Lipschitz extensions: minimizing the sup norm of the gradient*, Arch. Rational Mech. Anal. **123** (1993), 51–74. https://doi.org/10.1007/BF00386368.

[2] M. G. Crandall, L. C. Evans, and R. F. Gariepy, *Optimal Lipschitz extensions and the infinity Laplacian*, Calc. Var. Partial Differential Equations **13** (2001), 123–139. https://doi.org/10.1007/s005260000065.

[3] M. G. Crandall, H. Ishii, and P.-L. Lions, *User's guide to viscosity solutions of second order partial differential equations*, Bull. Amer. Math. Soc. (N.S.) **27** (1992), 1–67. https://arxiv.org/abs/math/9207212.

[4] C. Wang and Y. Yu, *C^1-boundary regularity of planar infinity harmonic functions*, Math. Res. Lett. **19** (2012), no. 4, 823–835. https://doi.org/10.4310/MRL.2012.v19.n4.a7.

[5] G. Hong, *Boundary differentiability for inhomogeneous infinity Laplace equations*, Electron. J. Differential Equations **2014** (2014), no. 72, 1–6. https://ejde.math.txstate.edu/Volumes/2014/72/hong.pdf.

[6] G. Hong, *Counterexample to C^1 boundary regularity of infinity harmonic functions*, Nonlinear Anal. **104** (2014), 120–123. https://doi.org/10.1016/j.na.2014.03.014.

[7] X. Feng and G. Hong, *Slope estimate and boundary differentiability for inhomogeneous infinity Laplace equation on convex domains*, Nonlinear Anal. **176** (2018), 36–47. https://doi.org/10.1016/j.na.2018.06.003.

[8] P. Juutinen, M. Parviainen, and J. D. Rossi, *Discontinuous gradient constraints and the infinity Laplacian*, Int. Math. Res. Not. IMRN **2016** (2016), no. 8, 2451–2492. https://doi.org/10.1093/imrn/rnv214. The patching statements used here are Section 2.2, Lemma 2.2 and Theorem 2.3 of https://arxiv.org/abs/1201.5199.

[9] OpenAI, *Uniform Interior $C^{1,\alpha}$ Estimates for Infinity-Harmonic Functions*, preprint, October 4, 2026. Preprint PDF at openai/math. The uniform interior estimate is used in the form of Theorem 1.1.